IMPROVING ACCESS TO UNTRANSCRIBED SPEECH CORPORA USING AI

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Abstract

How easily speech corpora can be indexed and searched has a direct impact on how effectively its contents can be used by many interested parties — from linguists, to language teachers, to community members. As transcribing speech is much more time consuming than recording it, large parts of speech corpora typically remain untranscribed, making it difficult to index and search these sub-parts. While searchable transcriptions can be automatically derived using a speech-to-text system for major languages like English, such technologies are typically unavailable for smaller languages, especially those typical in language documentation work. For documentation projects, this difficulty creates a bottleneck for creating language learning materials for language revitalisation and maintenance as well as linguistic analyses. In this dissertation, I propose four approaches to widen this bottleneck to enable some form of search or indexing, or accelerate the time-consuming process of transcription. Each chapter addresses a common but distinct scenario within language documentation projects according to the types and amounts of available data. For each scenario, I propose a context-appropriate, data-efficient solution that leverages AI speech models as well as external resources where appropriate.

Acknowledgments

As stated in the title page, this dissertation is only a *partial fulfilment* of the requirements — certainly a major step but nonetheless one of many in a long journey. I could not have completed this journey without the support of many within Stanford Linguistics and beyond who helped to make it a very fulfilling one.

First, I would like to thank those who not only made it possible for me to even start this journey but also be as prepared as possible for it. I thank Kate Lindsey for paving the way through Stanford Linguistics, exemplifying (and advising me on) ways of combining and complementing the resources and expertise of the department with those beyond — including a one-year deferral prior to starting the doctoral program. I thank Jane Simpson and Nick Thieberger for finding the funds to support me for an extra year at CoEDL (ARC Centre of Excellence for the Dynamics of Language).

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I'm grateful to Lily Clifford and Elisa Kreiss who made up our tiny cohort of three, sharing many of the excitements and stresses of first year including many early mornings (e.g. attending lexical semantics classes) and late nights (e.g. finishing lexical semantics homework). I'm also grateful that the start of my Stanford journey overlapped with the final years of Kate Lindsey, Simon Todd, and Rob Voigt each of whom was able to impart much wisdom before their departures. I thank Rob in particular for leading by example on how to gently but persuasively encourage participation in the department band. I hope I have imparted as much to the now band (co-)leaders: Irene Yi, Yin Lin Tan and Antón de la Fuente.

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up experiments. This series of e-mail discussions then led to a published paper, then several more (four in total), his short visit to Stanford (Fall quarter 2022) and now a longer one as a post-doc. I'm forever grateful to Martijn for the many brainstorming meetings, puzzling code debugging and intense last-minute writing marathons.

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Chapter 1

Introduction

> The major challenge for language documentation in the next decade or two is what could be called the transcription challenge. This is a multilayered challenge that goes far beyond the practical challenge of speeding up the transcription process. [...] Despite its centrality to language documentation, transcription remains critically undertheorized and understudied. Further progress in language documentation, and ultimately also its overall success, crucially depends on further investigating and understanding the transcription process, broadly conceived.
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> Himmelmann, N. P. (2018). Meeting the transcription challenge. In B. McDonnell, A. Berez-Kroeker & G. Holton (Eds.), *Reflections on language documentation: 20 years after Himmelmann 1998*.

Language documentation is typically considered a sub-field of linguistics and its goal is "to provide a comprehensive record of the linguistic practices characteristic of a given speech community" (Himmelmann, 1998, p. 166). Many aspects of its practice, however, require continual integration of developments beyond linguistics. As the target languages are often spoken by minoritised Indigenous communities, it is necessary to proactively consider evolving norms regarding fair and ethical collaboration (Holton et al., 2022). Concurrently, data protocols must not only continually integrate these norms but also technological advancements that more effectively enable the creation, processing, storage, and distribution of recorded materials (Berez-Kroeker et al., 2023). In both respective areas, there are ongoing developments that constitute paradigm shifts: 1) decolonising, community-centred approaches to working with Indigenous communities, and 2) speech and language processing systems powered by artificial intelligence (AI), specifically large-scale pre-trained models. Integrating considerations from the former and leveraging advancements in the latter, I examine in this dissertation how large-scale pre-trained models can be leveraged to develop speech processing systems to create contextually-appropriate solutions to combat a major bottleneck in language documentation workflows — the transcription of recorded materials.

1.1 Language documentation and the transcription bottleneck

The goal of language documentation is to produce a multi-purpose record of a language (Himmelmann, 1998). Given this goal, the recorded materials may encompass a variety of linguistic topics and cultural practices and, consequently, materials appropriate for one purpose (e.g. linguistic analyses) may be entirely inappropriate for others (e.g. creating language learning materials). Thus, the ability to efficiently locate and extract only the relevant information from recorded materials is a need shared among in a wide variety of use cases — from linguistic and lexicographical analyses, to development of language materials, to simply wanting to listen to the speech of one's elders. In this section, we examine how transcription plays a central role in meeting the various information needs for many diverse uses, how needs often go unmet due to difficulties of creating transcriptions (§1.1.2), and two pathways for better meeting these needs (§1.1.3).

1.1.1 Information needs met via transcription of recorded materials

It is generally recognised that there are about 7,000 languages spoken in the world today and that at least half of them may not exist by the end of the century (Austin & Sallabank, 2011). Many of these languages are spoken by Indigenous and minoritised communities, who are under cultural, economic, and technological pressures to shift to using more dominant regional, national, or global languages. For example, education may be conducted entirely in more widely-used regional or national languages and technological interfaces may only exist in a national or global language. Such pressures can lead to language endangerment, as younger generations gradually adopt the more widely-used languages until there remain no living first-language or 'L1' speakers, at which point a language may become 'dormant'. Thus, there are many ongoing revitalisation efforts to fight these pressures by encouraging the learning and active use of endangered languages, to safeguard the knowledge of elder L1 speakers by recording them, as well as efforts to 'awaken' dormant ones based on archival recordings.

Given a collection of recordings, the conventional workflow for making the collection searchable for various uses is to create time-aligned text annotations for each recording. As illustrated in Figure 1.1 using a Samoan utterance from my Field Methods class recordings, these annotations typically consist of a transcription of the utterance in either the target language orthography (if one exists) or in the International Phonetic Alphabet (IPA), a morpheme-by-morpheme gloss, and an utterance-level free translation of the utterance in a more widely-used vehicular language (e.g. English). These text annotations can be searched using plain text or regular expression queries. For a linguistic analysis, one may be interested in all forms with a glottal onset in the final syllable, which can be retrieved based on a regular expression search on the transcriptions (e.g. using the query '*ʔ[aeiou]+$'). For creating language learning materials illustrating verb tenses, one may want to find all instances of the verb [siʔi] 'to lift', which can be retrieved by either a search for 'siʔi' in the transcriptions or 'lift' in the glosses and/or translations. In both cases, a result of interest will be the instance occurring around the 30-second mark in the recording '20200218_TF_NS.wav' illustrated in Figure 1.1. In this way, these annotations play a central role in meeting a variety of information needs for many diverse uses.

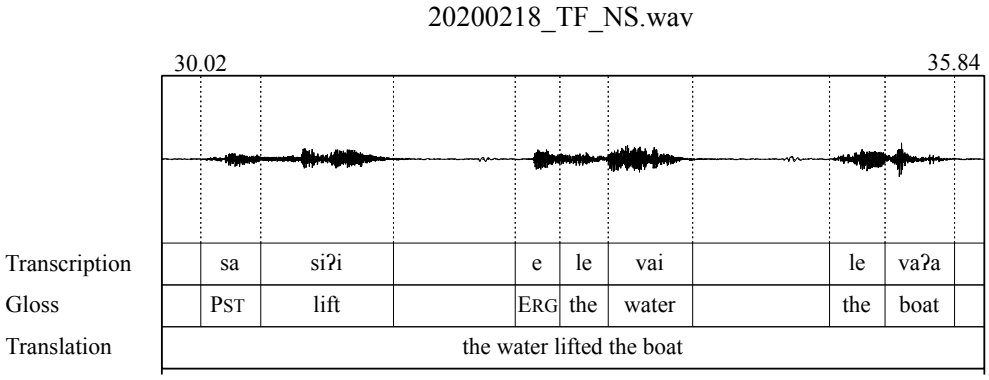


Figure 1.1: A Samoan utterance transcribed, glossed, and translated.

We note here two important details regarding information needs. First, we examine various information needs through various downstream 'uses' rather than 'users'. This is in recognition that linguists working in documentation are often required to 'wear many hats' (working on theoretical analyses at times and creating language learning materials at others), and that historical divides such as [non-native] linguist vs. [native speaker] community member are becoming increasingly tenuous given many instances of speaker-linguists working on their own languages. Second, we note that certain uses require considerably more detailed annotations than others. For example, morpheme-by-morpheme glosses would be essential for lexicographical analyses and the creation of a dictionary. By contrast, even partial translations may suffice for broadly identifying regions in the recordings where various topics of interest are being discussed. We return to this second fact below in examining how — for certain uses — some form of search and retrieval can be facilitated without the need for certain annotations that are time-consuming to create.

1.1.2 The transcription bottleneck: consequences and causes

As illustrated in Figure 1.2, the conventional workflow in language documentation projects consists of three broad stages: 1) collection, 2) annotation, and 3) use. As mentioned, the annotation stage (of which transcription is part) plays a central role in enabling audio recorded in the first stage to be searched and retrieved for various uses in later stages (e.g. for creating linguistic analyses or language learning materials). However, as manually transcribing recorded speech is much more time-consuming than recording speech, a bottleneck is formed at the transcription stage. As a consequence, many recordings within language documentation projects remain only partially transcribed, if at all (Cox et al., 2019). Yet, raw audio data comprising untranscribed speech is difficult to index and search, limiting how efficiently content within the materials can be discovered and used. Indeed, the consequences of hard-to-access materials are so dire that there are warnings against inadvertently creating "data graveyards, i.e. large heaps of data with little or no use to anyone" (Himmelmann, 2006, p. 4).

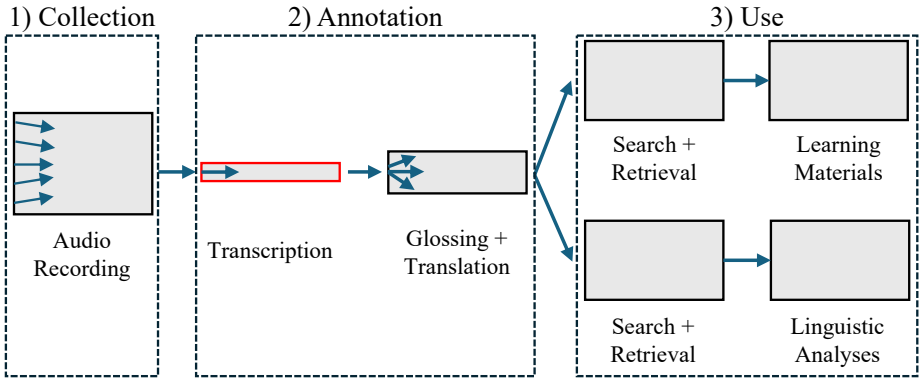


Figure 1.2: The transcription bottleneck within the standard workflow for processing and using audio recordings collected for language documentation, adapted from (Bird, 2020, Figure 1).

There are several contributing factors that result in this bottleneck with respect to transcribing minoritised languages. First, unlike major languages such as English, searchable high-quality transcriptions cannot simply be derived using a high-performance automatic speech recognition (ASR) system, whose development itself has conventionally required large quantities of transcribed speech. Second, there is typically no option for large-scale crowd-sourcing of target language transcriptions as by definition of endangerment there may be very few speakers and only a subset of speakers may be literate in this language (i.e. not the regional or national language of formal education). Third, there may not be a standardised orthography for the target language, requiring transcription in IPA or development of a project-specific working orthography, with both options resulting in transcriptions with both intra- and inter-transcriber variation, requiring additional time for review and corrections. Finally, in some instances, there may also be an additional personnel bottleneck resulting from limitations on who can listen to and transcribe certain recordings (e.g. of culturally-sensitive materials such as descriptions of ceremonial procedures). Taken altogether, surveys of transcription time have reported typically requiring about 30 to 50 hours of work to transcribe one hour of recorded speech (Do et al., 2014; Foley et al., 2019; Zahrer et al., 2020).

Reviewing a century of transcription practice within linguistics, Bird (2020) argues that some of these contributing factors result from a description-centred transcription practice, which we detail below. For context, we note here that language documentation and description are considered two inter-dependent but different activities varying in outputs and goals (Himmelmann, 1998). Language documentation, which includes data collection and annotation, aims to produce a multi-purpose record of a language. Its outputs are primary data that may be useful to a broad audience, including community members, language teachers, anthropologists, and those from other disciplines. The outputs of language description are typically grammatical descriptions and theoretical analyses based on these descriptions that are more narrowly aimed at comparative linguists. Historically, documentation processes have privileged the latter, which can result in primary data that is sub-optimal or inapplicable for other uses. For example, description-focused data collection may lead to the collection of idealised forms of infrequent phenomena-of-interest rather than day-to-day colloquial or

child-directed speech (Rouvier et al., 2017), the latter two of which are more relevant to revitalisation efforts. Spurred on in part by broader decolonial efforts by Indigenous communities, community-based and collaborative documentation methods that also contribute to community needs are emerging as best practices in linguistic research (Holton et al., 2022).

Bird (2020) argues that several parts of the transcription bottleneck result from a description-centred transcription practice. He identifies various inefficient characteristics particular to orthodox transcription practice, two of which are 'transcribing fully' and 'transcribing first' (i.e. translating second). We review these characteristics using the aforementioned Samoan utterance from my Field Methods class recordings, repeated below for convenience as Figure 1.3. Given my goal of grammatically analysing a language largely unknown to me, I am primarily interested in textually representing the form of the spoken utterance, leading to a primary focus on the phonetic form of the whole utterance [sasiʔielevaileva ʔa], with the translation secondary. Using these annotations along with other data points, I can hypothesise what the various lexical and grammatical units are in Samoan and how they interact, e.g. [sa]+[siʔi] PST+lift 'lifted'. Bird (2020) questions whether this description- and analysis-centred conceptualisation of transcription effectively serves the many parties interested in working with untranscribed language documentation corpora, particularly those of unwritten, oral languages.

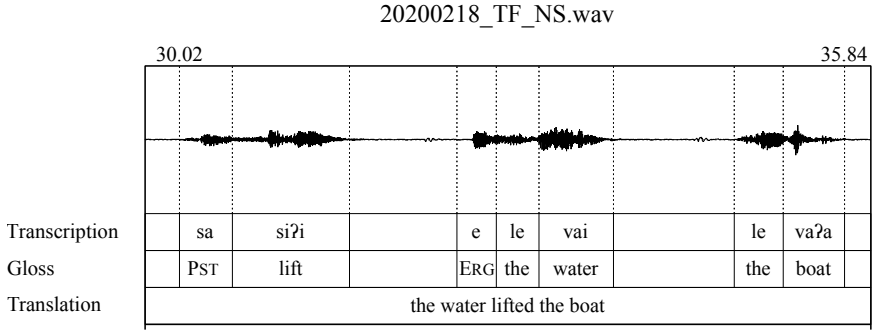


Figure 1.3: A Samoan utterance fully transcribed, glossed and translated for linguistic analysis.

1.1.3 Two pathways for alleviating the transcription bottleneck

As noted above, not all uses require fully transcribed, glossed, and translated recordings for discovering excerpts of interest within documentation corpora. For example, a community member may simply wish to listen to excerpts of a certain elder or all excerpts relating to a broad topic (e.g. place names). Often, recording-level metadata and/or accompanying field notes or wordlists can help offer a rough index of the recordings, though such data sources may not always be available — especially in the case of archival recordings. Indeed, as originally pointed out by Bird (2020), there is little reason for search and retrieval for these uses to await for the corpus to be fully transcribed or even the development of a target language orthography. Consequently, one important research question (RQ1) for improving access is how to derive a rough but workable index of the recordings that bypasses target language transcription — avoiding the transcription bottleneck altogether.

Nevertheless, certain uses do require excerpts of interest to be fully transcribed, glossed, and translated. For example, creating a dictionary of the language typically requires some linguistic and lexicographic analyses to catalogue the lexical and grammatical units in the excerpts, e.g. [sa]+[si?i] PST+lift 'lifted'. For uses that depend on prior analyses, the process of transcription remains an unavoidable part of the workflow. One natural solution is of course to use automatic speech recognition (ASR) to derive first-pass transcriptions that can be corrected by a human, thereby reducing transcription time (Prud'hommeaux et al., 2021; Sperber et al., 2016, 2017). As mentioned, however, developing an ASR system has conventionally required a large quantity of transcribed speech which is hard to obtain — leaving many minoritised languages in a Catch-22. Consequently, a second important research question (RQ2) is how ASR systems can be developed using only limited amounts of transcribed speech by leveraging supplemental resources that are relatively easier to obtain (e.g. text from books or websites, untranscribed speech from news broadcasts).

At a broader level, we may view solutions that address each of these two research questions (RQ1, RQ2) as use-informed pathways that, respectively, bypass or widen the transcription bottleneck, as illustrated in Figure 1.4. For uses that do not require detailed analyses of the spoken content, the first pathway helps provide a rough but workable index with approximate locations of key words and phrases of interest in a more widely spoken vehicular language (e.g. English). For uses that do require detailed analyses and hence transcriptions of the spoken content, the second pathway helps accelerate transcription efforts by leveraging an ASR system to create rough first-pass transcriptions that can be corrected by a human transcriber. In this way, these pathways help diversify the avenues through which diverse information needs can be met.

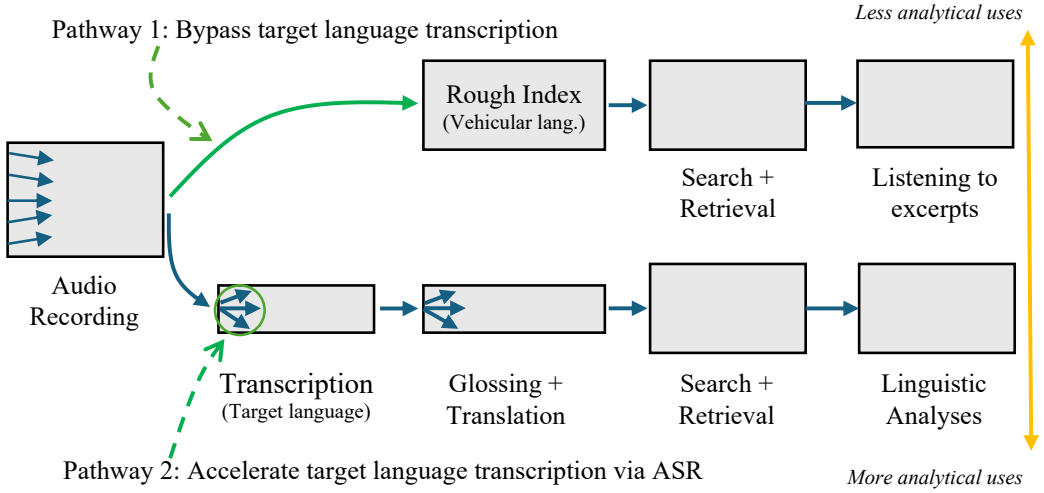


Figure 1.4: Two pathways for alleviating the transcription bottleneck: (1) Avoid the target language transcription bottleneck by deriving a rough index in a more widely-used vehicular language, and/or (2) Widening the bottleneck by using ASR to accelerate transcription.

1.2 Alleviating the transcription bottleneck using pre-trained models

In the previous section, I substantiated the importance of addressing two research questions related to alleviating the transcription bottleneck in language documentation projects. These questions were: RQ1) how to bypass the need for target language transcription and derive a searchable index in a more widely used vehicular language, and RQ2) how to make effective use of limited, hard-to-obtain target language transcriptions for ASR development by leveraging relatively easier-to-obtain supplemental resources. As mentioned, solutions that address each of these research questions can be broadly viewed as diversifying the pathways that can help meet diverse information needs. Given a diversity of scenarios in the kinds and amounts of available data from one documentation project to another, I propose for each pathway two possible implementations that each address a common but distinct scenario.

In this section, I provide high-level summaries of my implementations and key findings. For these high-level summaries, I will refer to recently developed artificial intelligence (AI) models used in the various implementations collectively as 'pre-trained models', a topic which will be discussed in detail in Chapter 2. Then, in Chapters 3–6, I detail each implementation addressing a specific scenario and one of the two research questions, as summarised in Table 1.1. In Chapters 3–4, I propose two implementations for deriving a rough but workable index in a vehicular language in scenarios where no target transcriptions are obtainable (e.g. due to a lack of a target language orthography). In Chapters 5–6, I propose two implementations for ASR-assisted transcription in scenarios where a small amount of target language transcriptions are available. In particular, I examine how to improve ASR system performance for delivering better quality first-pass transcriptions by drawing on supplemental resources that are relatively easier to obtain than transcribed speech (e.g. texts from books or untranscribed speech).

Chapter	Transcriptions (target lang.)	Corpus type	Other resources
RQ1: *How to derive rough index of untranscribed corpora in vehicular language?*			
3	-	Monolingual (target language)	-
4	-	Mixed (target + vehicular language)	SLI model & Vehicular language ASR
RQ2: *How to make best use of limited target language transcriptions using supplemental resources?*			
5	10 min.	Monolingual (target language)	Target language texts
6	1 hr	Monolingual (target language)	Untranscribed speech from similar language

Table 1.1: Language documentation scenarios according to amount of available target language transcriptions and composition of speech corpora addressed in the different chapters in the dissertation. Initialisms: Spoken Language Identification (SLI), Automatic Speech Recognition (ASR).

1.2.1 Deriving a rough index in a vehicular language using pre-trained models

As discussed in the previous section (§1.1.2), Bird (2020) identifies two attributes in the conventional annotation workflow that contribute heavily to the transcription bottleneck: transcribing fully and transcribing first (i.e. translating second). While these attributes are well-motivated for certain uses (e.g. linguistic description), not all uses require such in-depth and thus time-consuming analyses of the spoken content. To facilitate search and retrieval for these latter uses, Bird (2020) suggests relaxing these two attributes by, respectively: annotating sparsely (instead of fully) and using translations (instead of target language transcriptions).

As illustrated in Figure 1.5, Bird (2020) proposes a supplementary annotation workflow where spoken term detection or 'word spotting' is used to facilitate searches via spoken queries (i.e. a voice-based alternative to text queries on transcriptions) and, upon confirmation of hits, time-aligned translations can be provided in a vehicular language (i.e. one with a standardised orthography in which speakers are literate). The confirmed locations of various queries along with their translations allow for the corpus to be collaboratively and iteratively indexed, albeit sparsely (i.e. at the outset, just key words of interest such as common verbs and nouns, as illustrated in Figure 1.5). Additionally, this approach can help more effectively find regions of interest to which more time-consuming conventional transcription efforts can be strategically directed (i.e. it is intended as a supplementary not alternative approach to 'full' transcription).

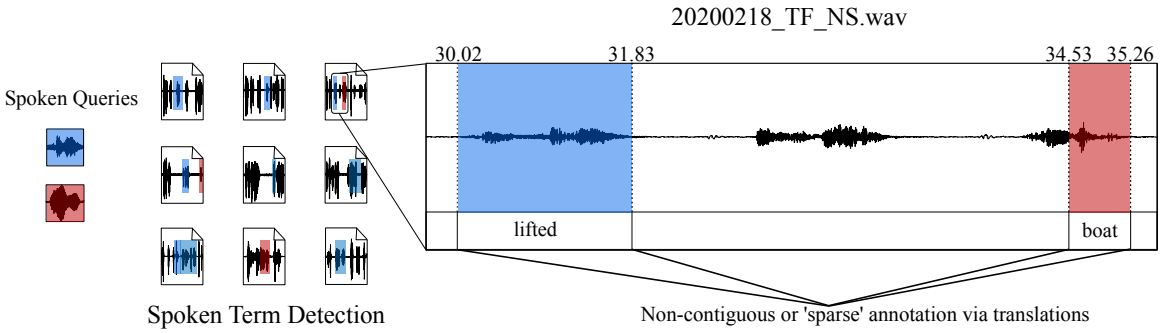


Figure 1.5: Sparse annotation via translations in a vehicular language. Spoken term detection is used to enable search and retrieval using voice queries.

Building on this approach, I suggest an implementation for deriving a rough index in a vehicular language without spoken term detection or manual crowd-sourced translations for a specific type of speech corpus commonly found in documentation projects. In language documentation, recordings may contain not only the target language being documented but also vehicular language used for questions and commentary about the target language, i.e. the vehicular language (e.g. English) is used as a metalanguage. For example, as illustrated in Figure 1.6, the target language response by the Samoan speaker around the 30-second mark was preceded by my elicitation prompt in English: *How do you say "the water lifted the boat"?* As mentioned above, it is relatively straightforward to derive searchable high-quality transcriptions using a high-performance ASR system for major languages such as English. As such, I show that for this type of mixed-language corpora

pre-trained models can be used to automatically isolate and transcribe the metalanguage. These searchable metalanguage transcriptions in turn provide approximate locations where certain target language words and topics are being discussed.

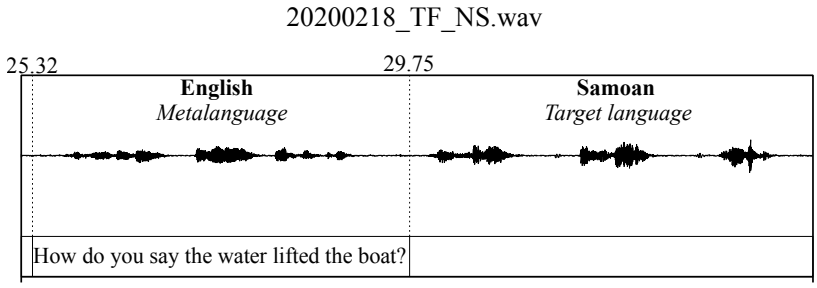


Figure 1.6: Deriving a proxy index via automatic isolation and machine transcription of a more widely used vehicular language in a mixed-language corpus.

Chapter 3: Improving spoken term detection using pre-trained models

As introduced above, spoken term detection could potentially make an untranscribed speech corpus more accessible by allowing users to perform voice searches and in turn permit confirmed hits to act a sparse index of the corpus. Recent work investigating this potential discovered that while retrieval performance was tolerable in cases where the speakers of the search query and the searched recordings were the same, much worse performance was observed when the two were different (Le Ferrand et al., 2020). Based on their comparison of various speech representations, Le Ferrand et al. (2020) suggested future work investigate whether recently developed pre-trained models offer more speaker-independent representations.

Accordingly, I compared the impact on retrieval performance as a result of using speech representations extracted from various pre-trained models as well as those from two baseline methods, using data from 7 Australian Aboriginal languages and Gronings, a variant of Low Saxon (one of the minority languages spoken in the Netherlands).

Results from these experiments showed that in the worst-case scenario retrieval performance is still improved 56–86% from only 27–28% of queries being retrievable for the best baseline to 42–52% when using pre-trained models. Given the alternative is browsing the untranscribed audio in near real-time, this level of performance helps enable *some* form of search and retrieval to untranscribed speech corpora.

Chapter 4: Deriving a metalanguage proxy index using pre-trained models

As introduced in the previous section (§1.1.2), there may be access restrictions to recordings in a documentation project as a result of potential culturally-sensitive materials within these recordings. However, for archival materials with insufficient metadata, it is hard to determine which parts of which recordings must be restricted without the custodian(s) manually listening to and vetting each recording in the collection.

I propose a privacy-preserving workflow to accelerate this vetting process for recordings where speech in the target language is intermixed with a more widely-used language such as English for meta-linguistic commentary and questions (e.g. *What is the word for 'tree'?*). I propose integrating voice activity detection (VAD), spoken language identification (SLI), and automatic speech recognition (ASR) to isolate and transcribe the metalinguistic content, which an authorised person can quickly scan to triage recordings that can be annotated by people with lower levels of access.

I report results from an ongoing documentation project to process 136 hours archival audio containing a mix of English and Muruwari. In this collaborative work with the Muruwari custodian of the archival materials and the documentation team, I show that this workflow reduces metalanguage transcription time by 20% even with minimal amounts of annotated training data to adapt pre-existing task-specific models: as little as 10 utterances per language for SLI and 39 seconds for English ASR. These metalanguage transcriptions not only allow the custodian to more efficiently vet and clear recordings (or hold certain ones back for further manual review), but also allow the documentation team to search the cleared materials for creating language revitalisation materials and linguistic analyses.

1.2.2 ASR development with limited target language transcriptions leveraging pre-trained models and supplemental resources

As discussed in the previous section (§1.1.3), there are many uses which require in-depth analysis of the spoken language content and hence transcription of this content. However, manual transcription of recordings in documentation projects can require between 30 to 50 hours of work to transcribe one hour of recorded speech (Do et al., 2014; Foley et al., 2019; Zahrer et al., 2020). While there has been a long-standing desire to accelerate the transcription process using automatic speech recognition (ASR), such systems have typically been out of reach for many documentation projects — given large quantities of transcribed speech have typically been required to begin ASR system development.

Recently developed pre-trained models, however, have substantially reduced this upfront cost, demonstrating that ASR system development may now be possible with even an hour of transcribed speech or, in certain circumstances, as little as 10 minutes. Naturally, this has spurred on much experimentation for leveraging such models for ASR-assisted transcription in language documentation workflows (e.g. Bartelds et al., 2023; Coto-Solano et al., 2022; Guillaume et al., 2022; Macaire et al., 2022). Understandably, given that transcribed speech is *the* critical resource for ASR development, many of these investigations have focused on understanding the impacts on ASR system performance when using pre-trained models with various amounts of transcribed speech. Complementing these investigations, I examined how to make most effective use of a fixed amount of transcribed speech by drawing on other easier-to-obtain resources.

In the broader ASR literature, 'low resource' languages are typically classified as such primarily based on the availability of transcribed speech. Other resources and information for these languages are assumed to be readily obtainable. For example, it is assumed that text data is readily available from the web (e.g. Wikipedia), untranscribed speech from radio or news broadcasts, or reliable metadata (e.g. geographic location,

genetic family) is available in typological databases. Such assumptions, however, may not hold true for under-described languages with little digital representation typical in language documentation projects. Accordingly, in my investigations presented in Chapters 5–6, I investigated when certain proposals within the low-resource ASR literature were directly applicable for use in documentation projects and possible alternative pathways for when they were not.

Chapter 5: Kick-starting ASR development using pre-trained models and supplemental texts

As mentioned, recent experiments have demonstrated that ASR development may be possible with as little as 10 minutes of transcribed speech when leveraging pre-trained models and some supplemental text data. The text data can be leveraged to create a lexicon that improves the performance of an ASR system by limiting its output to possible words (e.g. **dogz* vs. *dogs*) and biasing the system towards probable word sequences (e.g. *too dogs* vs. *two dogs*). These experiments, however, leveraged a pre-trained model developed using 960 hours of English audio and, crucially, an 803 million word corpus of English text. Yet, language-specific pre-trained models and this much text data are unlikely to be available for minoritised languages.

I examined the real-world viability of this setup beyond English using a multilingual pre-trained model and conducted ASR experiments using 10 minutes of transcribed speech from two pairs of languages. These languages differed in the availability of supplemental text data: Gronings and Frisian (two West-Germanic languages for which ~7.5M word corpora available) and Besemah and Nasal (two Malayo-Polynesian languages for which only small ~10k word lists are available).

Results showed that this setup does appear promising for the two West-Germanic languages for which sufficient text data can be sourced and are over-represented in the training data on which the multilingual pre-trained model was developed — yielding an average word-error rate of 39% with only 10 minutes of transcribed speech and as little as an 80k word corpus of text (i.e. roughly the length of a novel). However, for the two under-represented, predominantly oral languages for which little text data is available, the ASR performance remained poor (65% average) — suggesting the need for a different setup for such languages.

Chapter 6: Leveraging untranscribed speech to adapt pre-trained models for under-represented languages

As European languages dominate the training data for multilingual pre-trained models, directly using the off-the-shelf pre-trained model for ASR for under-represented languages results in relatively lower performance compared to their well-represented counterparts (Conneau et al., 2023). While recent studies have shown that these models can be adapted using untranscribed speech from the target language in turn closing this performance gap (e.g. Nowakowski et al., 2023; Paraskevopoulos et al., 2024), they involved using 70–200 hours of target language data. For some languages, it may be quite difficult to source this much speech data — even untranscribed.

For these languages, I propose carrying out the model adaptation process by supplementing limited amounts of target language data with data from a similar, higher-resource language (termed the 'donor' language). Additionally, I suggest that the acoustic-phonetic and phonotactic similarities between the target and donor languages predict improved ASR performance in the target language and propose a bottom-up measure to determine this similarity — the Acoustic Token Distribution Similarity (ATDS).

Results showed improved ASR performance in the target language (e.g. Punjabi) when models were adapted using supplemental data from a similar donor (e.g. 10h Punjabi + 60h Hindi) but not when using data from a dissimilar one (e.g. 10h Punjabi + 60h Bengali). Across a set of typologically different target languages (Punjabi, Galician, Iban, Setswana), I show that the ATDS between the target language and its candidate donors precisely predicts target language ASR performance and outperforms all other similarity measures examined.

1.3 Published papers and multiple authorship statement

The work presented in this dissertation is the result of collaborations with other researchers and has already been published or accepted for publication at the time of writing. In this section, I provide statements about these publications with multiple authorship for each chapter, outlining my personal contributions.

1.3.1 Chapter 2

Chapter 2 is a published research paper (San, Bartelds et al., 2021), completed under the mentorship of my advisor Dan Jurafsky. This paper arose out of my second qualifying paper (QP2) on the same research topic (spoken term detection). I expanded upon my QP2 work by adding additional experiments with the help of collaborator (and co-first author) Martijn Bartelds as well as additional datasets on which these experiments were conducted. Mitch Browne, Fiona Gibson, John Mansfield, David Nash, Jane Simpson, Myfany Turpin, Maria Vollmer, and Sasha Wilmoth contributed linguistic fieldwork data to these experiments. All experiment code and paper was co-written with Martijn Bartelds. Experiment design was my own, based on my QP2 work.

1.3.2 Chapter 3

Chapter 3 is a published research paper (San et al., 2022), completed under the mentorship of my advisor Dan Jurafsky. The paper proposes a data processing workflow for recordings of Muruwari, an Australian Aboriginal language. Alison Mount, Ruben Thompson, Michael Higgins, Roy Barker, and Jane Simpson comprise the Muruwari language documentation team. Tolúlọpẹ́ Ògúnrẹ̀mí and Martijn Bartelds assisted with experiment implementation. Experiment design and writing are all my own.

1.3.3 Chapter 4

Chapter 4 is a published research paper (San et al., 2023), completed under the mentorship of my advisor Dan Jurafsky. The paper evaluates a proposed workflow using four languages, two of which are Besemah and Nasal (two Austronesian languages spoken in Indonesia). Blaine Billings, Ella De Falco, and Bradley McDonnell comprise the Besemah and Nasal language documentation team, along with community members Hendi Feriza (Besemah), and Johan Safri and Wawan Sahrozi (Nasal). Ben Foley and Martijn Bartelds assisted with experiment design. All experiment code and writing are my own.

1.3.4 Chapter 5

Chapter 5 is a paper accepted for publication at the time of writing (San et al., to appear), completed under the mentorship of my advisor Dan Jurafsky. Georgios Paraskevopoulos and Oliver Adams assisted with experiment design. Xiluo He assisted with running experiments. Punjabi speaker/language experts Aryaman Arora and Prabhjot Kaur assisted with hypothesis refinement and error analyses. All experiment code and writing are my own.

Chapter 2

Speech processing using pre-trained models

As outlined at the end of the previous chapter (§1.3), the work presented in this dissertation has been published or accepted for publication at the time of writing (March 2024), with some appearing several years earlier (e.g. San et al., 2021). That is, some areas have seen even more recent developments following the work presented. The goal of this chapter is thus twofold: to not only provide the background on speech processing and recently developed pre-trained models to understand content contained in this dissertation, but also situate the content in relation to related work current at the time of writing. Additionally, to cater to multiple audiences and interests, from the documentarian wanting a broad overview of speech processing to a computational linguist wanting to build on this body of work, this chapter is divided into three parts, each increasing in the level of technical detail. In the first part (§2.1), I provide a high-level introduction to speech processing and various technical terms used in the remainder of this dissertation. In the second part (§2.2), I provide a mixed-level overview of wav2vec 2.0 (a framework for developing pre-trained models for speech processing), where occasional low-level details are provided as footnotes to relatively high-level text. In explicit terms: the start and ends of each paragraph in this section provide high-level explanations along with narrative scaffolding, though lower-level details may also be present in the middle of some. Finally, in the third part (§2.3), I provide a synthesis of related work where I assume some familiarity with the lower-level details of various models and methods. The remainder of the dissertation may be understood without reading this section.

2.1 An introduction to speech processing

2.1.1 Speech processing tasks

The various tasks found in the speech processing literature can be understood as processes that help annotate speech with specific information. Figure 2.1 provides a summary of common speech processing tasks, illustrated using an excerpt of speech where an English speaker is eliciting from a French speaker some word of

											
Time window	1	2	3	4	5	6	7	8	9	10	11
Voice Activity Detection *Is there speech?* Y[es], N[o]	N	Y	Y	Y	Y	N	N	Y	Y	Y	N
Speaker Diarisation *Who is speaking?* Speaker A or B		A	A	A	A			B	B	B	
Spoken Language Identification Which language? E[NG], F[RA]		E	E	E	E			F	F	F	
Automatic Speech Recognition (ENG) *What is being said?* a, b, c, …, z		s	o	u	l						
Automatic Speech Recognition (FRA) *What is being said?* a, â, à, …, z								â	m	e	

Figure 2.1: A high-level illustration of various speech processing tasks, illustrated using an English (ENG) speaker eliciting a word of interest from a speaker of French (FRA).

interest, *âme* 'soul'. At a high-level, each of these tasks can be understood as training a model to predict the answer to a multiple-choice question for a given time window of speech. Each task differs from the others in the question under consideration and in the possible answers, e.g. Voice Activity Detection (VAD): *Is there speech?* Yes/No. Despite these differences, all tasks have a common need to extract relevant information from the speech signal. In other words, all tasks stand to benefit from an adaptable, general-purpose model that can assist in this common sub-task of information extraction.

2.1.2 Speech processing using pre-trained models

An emerging class of models for building artificial intelligence (AI) systems is foundation models (FMs: Bommasani et al., 2021), also known as 'general-purpose' or 'base' or 'pre-trained' models.[1] Across many domains, these models have led to a paradigm shift in AI system development. As illustrated in Figure 2.2, this paradigm consists of a pre-training stage where a model is first trained on unlabelled data using a proxy task which permits the resulting pre-trained model to be re-used for various target tasks at a later stage. A common proxy task for pre-training is masked modelling where, as illustrated in Figure 2.2 (i), the model is given a partially masked input and is then tasked with using the neighbouring, unmasked regions to predict the contents of the masked ones. As labels for this proxy task can be derived automatically without the need for humans (i.e. it is 'self-supervised'), the pre-training stage can leverage substantially larger amounts of data than previously possible (e.g. half a million hours of untranscribed speech). This large-scale pre-training is commonly considered a key ingredient for the adaptability of these models for various downstream tasks even with only limited task-specific data (Bommasani et al., 2021).

[1] Given its widespread use in the speech processing domain, I adopt the term pre-trained models throughout this dissertation.

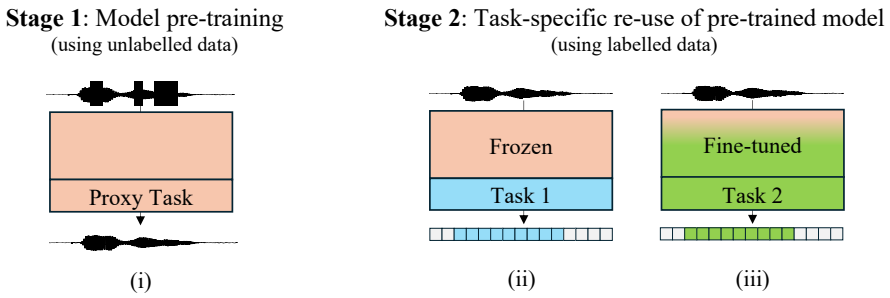


Figure 2.2: System development paradigm based on pre-trained models.

Given a pre-trained model, it may be re-used for target tasks in several ways. For relatively straightforward classification tasks (e.g. spoken language identification, spoken term detection), the information extracted from the input by the pre-trained model (or its 'representations') may be sufficient to use as-is. In these cases where the model is not further trained, or used 'frozen' as illustrated in Figure 2.2 (ii), its representations are simply fed into another model which is trained to predict the task-specific labels based on these representations. For more complex text-generation tasks (e.g. ASR), the model may need to be adapted for the target task through further training, or 'fine-tuning' as illustrated in Figure 2.2 (iii). In these cases, both the pre-trained and task-specific models are trained together to predict the task-specific labels. Additionally, if the data on which the model was pre-trained is significantly different from the data to be used in the target task, the pre-trained model can also be adapted via 'continued pre-training' (CPT: Gururangan et al., 2020) using all available unlabelled target data (e.g. untranscribed audio), only after which this adapted model is fine-tuned with the more limited labelled subset (e.g. audio paired with human transcriptions for ASR).

2.2 wav2vec 2.0: A framework for pre-training models for speech processing

In this section, I provide a detailed introduction to wav2vec 2.0, a framework for developing pre-trained models for speech processing that has seen widespread proliferation. As mentioned above, self-supervised pre-training has enabled models to be trained without requiring any human labels and consequently at increasingly larger scales. Since the release of the original English model pre-trained on 960 hours of English (Baevski et al., 2020), there have been additional larger-scale variants both domain-specific and multilingual, one or more of which were used for experiments presented in Chapters 3–6. For example, the Robust variant (Hsu, Sriram et al., 2021), intended for noisy speech domains, is pre-trained on 63k hours of English speech (including audio from noisy telephone conversations). Multilingual variants include XLSR-53, pre-trained on 56k hours from 53 languages (Conneau et al., 2021); XLSR-128, pre-trained on 436k hours from 128 languages (Babu et al., 2022); and MMS, pre-trained on 491k hours from 1,406 languages (Pratap et al., 2023).

At a high-level, the wav2vec 2.0 model architecture and pre-training process can be thought of as tasking the model to discover — based on raw audio alone — a useful set of discrete units (e.g. {*a, b, ..., z*}) with which it is able to carry out a speech variant of a Cloze test: e.g. given "*The big _ _ _ chased the small rat.*", correctly guess for each masked position *c*, *a*, and *t*, respectively. Accordingly, as illustrated in Figure 2.3, the wav2vec 2.0 architecture consists of three parts: 1) a convolutional feature extractor that extracts learnable features from strided frames in the audio input, 2) a quantiser which clusters the audio features into a set of discrete code vectors (q_1, ..., q_5), and 3) a transformer attention network that learns context-enriched representations (h_1, ..., h_5). For self-supervised pre-training, the model is optimised using a joint contrastive loss and a diversity loss. The contrastive loss requires the model to use the neighbouring context to distinguish masked frames amongst a set of negative distractors. The diversity loss requires the model to make equal use of all code vectors, preventing the model from relying on a small subset which can lead to trivial/sub-optimal solutions.

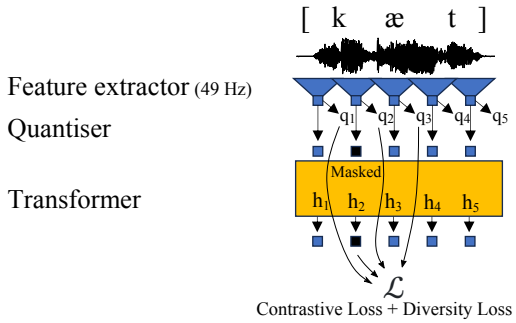


Figure 2.3: Illustration of the wav2vec 2.0 architecture. Adapted from Baevski et al. (2020).

One consequence of the wav2vec 2.0 model architecture and pre-training objectives is that the representations learned through pre-training are relatively fine-grained. For example, as illustrated in Figure 2.3, a single speech sound such as [æ] may last for three time steps (yielding three code vectors: q_2, q_3, q_4), since the model encodes speech at a rate of of 49 Hz. When combined with the diversity loss that requires the model to equally make use of all code vectors, this results in a complex many-to-many mapping between the learned discrete codes and more coarser linguistic units such as phones.

2.2.1 Probing wav2vec 2.0 representations

One active area of research is in understanding what kinds of information are learned at the various stages in the models (e.g. Abdullah et al., 2023; Bartelds et al., 2022; Ma et al., 2021; Pasad et al., 2021, 2023, 2024). One finding from this body of work is that, while the mapping between them is complex, the discrete codes do consistently correspond to a specific of speech sound, or a set thereof. For example, as illustrated in Figure 2.4, each discrete code listed with an arbitrary alphanumeric label on the x-axis (e.g. A46, A29, ..., B1) is typically associated with a small set of phones (e.g. A46 with velar stops [k] and [g]).

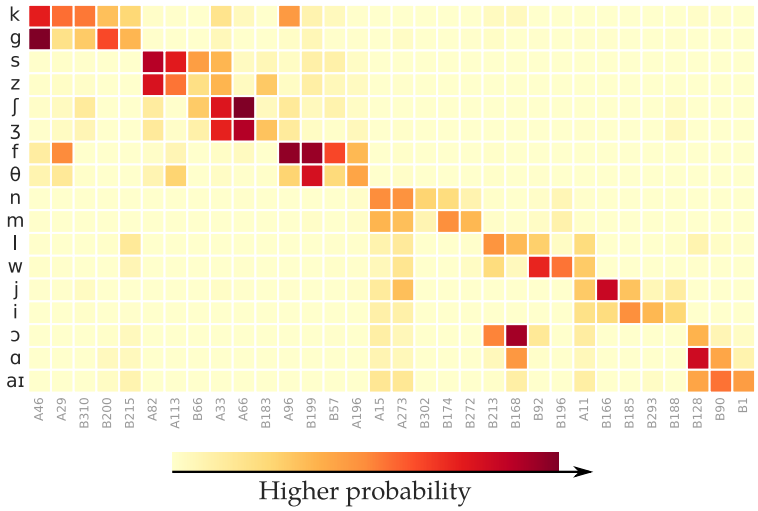


Figure 2.4: Probability of English phone categories for a given discrete unit learned by the wav2vec 2.0 XLSR-53 model. Figure sourced from Abdullah et al. (2023), used with permission from Badr Abdullah.

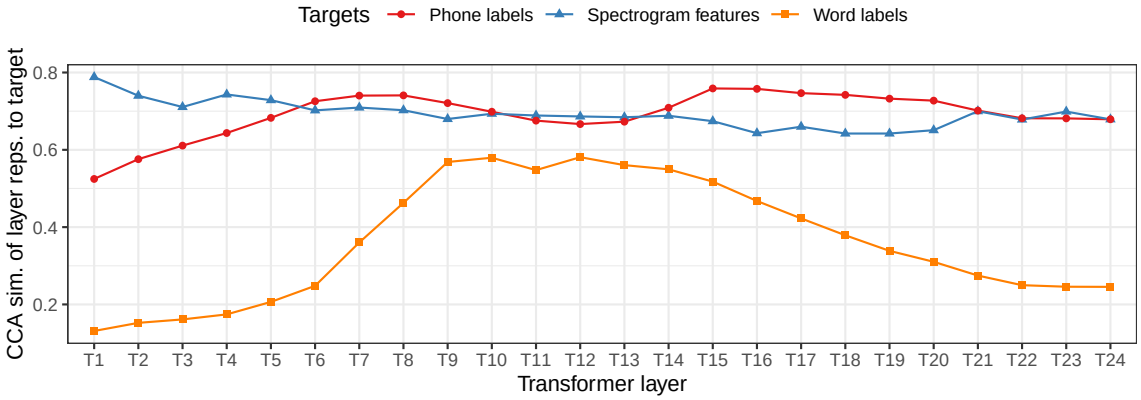


Figure 2.5: Similarity of transformer layer outputs to spectrogram features, phone labels, and word labels. Similarities calculated using canonical correlation analysis (CCA) on English using the wav2vec 2.0 LV-60 model, reported in Pasad et al. (2023). Figure re-produced using data provided by Ankita Pasad.

Another finding is that the type and degree of linguistic information contained in the model's intermediate representations varies from layer to layer. For example, as illustrated in Figure 2.5, the representations of the very first transformer layer (T1) is highly similar to those extractable using conventional spectrograms and this similarity decreases in later layers. Similarly, linguistic information related to word identity is found in the middle layers (e.g. T12) while information related to phone identity in later ones (e.g. T16). Investigating what information is learned in which layers and why is an active area of research, only briefly noted here. For our interest in using wav2vec 2.0 models for various speech tasks, we simply note that certain tasks may benefit more from using a middle layer while others benefit from later ones.

As mentioned above, a pre-trained model such as wav2vec 2.0 may be used 'frozen' (i.e. without additional training to simply help extract speech information) or 'fine-tuned' (i.e. with additional training using task-specific labels such as transcriptions for the ASR task). In the remainder of this subsection, I detail the use of wav2vec 2.0 models in each of these two configurations through two relevant tasks: spoken term detection and automatic speech recognition.

2.2.2 Using frozen wav2vec 2.0 representations for spoken term detection

In the task of spoken term detection, we are interested in determining whether a given spoken query occurs within some other spoken utterance, known as the 'test item'. Given that a query may be spoken by one person in one recording environment (e.g. a loud, high-pitched speaker in a quiet room) while the test item may be spoken by another person in a different one (e.g. a quiet, low-pitched speaker in noisy room), a critical part of successfully carrying out this audio-on-audio comparison is obtaining a speaker-invariant and noise-robust representation of the phonetic content in the query and test item.

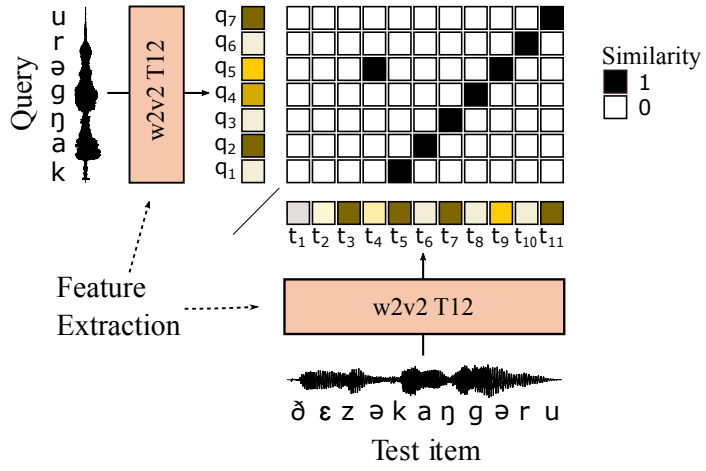


Figure 2.6: Using transformer representations from a wav2vec 2.0 model as a features for the spoken term detection task.

The use of a wav2vec 2.0 model as a feature extractor for spoken term detection is illustrated in Figure 2.3. In this example, we have a query *kangaroo* [kaŋgəru] and test item *there's a kangaroo* [ðɛzəkaŋgəru], and are interested in correctly detecting that the query does indeed occur within the test item. To carry out this task, we first derive speech representations extracted from the 12th transformer layer of a wav2vec 2.0 model, which we will see in Chapter 3 is consistently the most well suited layer for this task. This feature extraction process yields for the query a sequence of vectors $q_1, ..., q_7$ each representing the speech content at some given time window in the query. Similarly, we use the same process to derive a corresponding sequence for the test item, $t_1, ..., t_{11}$.

These vectors can then be used by a spoken term detection model which compares the similarity of each pair of vectors (e.g. q_1 vs. t_1, ..., q_7 vs. t_{11}) and returns a score of how likely the query is to be found within the test item. For example, q_1 vs. t_1 has a similarity of 0, given the [k]-initial query ***k**angaroo* vs. the [ð]-initial test item ***th**ere's a kangaroo*. By contrast, q_7 vs. t_{11} has a similarity of 1 as both end in the same vowel [u]. Conventionally, this detection stage is typically carried using using some form of Dynamic Time Warping (DTW) based algorithm, though we review more recent alternatives in the related work section of this chapter. Roughly summarised, the task of this model is to use the representations to look for whether there is a highly correlated region between the query and test item, as illustrated by the diagonal in Figure 2.6.

2.2.3 Fine-tuning wav2vec 2.0 models for automatic speech recognition

As mentioned, in more complex text generation tasks such as ASR, the wav2vec 2.0 model is typically further trained or 'fine-tuned' for this target task. This is typically done by appending another model,[2] after which both models are jointly optimised for the ASR task.[3] Similar to the frozen use, this process amounts to using the pre-trained wav2vec 2.0 model to encode the speech into some intermediate set of representations from which a decoding process uses those representations to generate a sequence of characters, e.g. *too dogz* as illustrated in Figure 2.7. In contrast to the frozen use case,[4] the fine-tuning process adjusts the wav2vec 2.0 model to better extract representations optimised for this new text generation task.

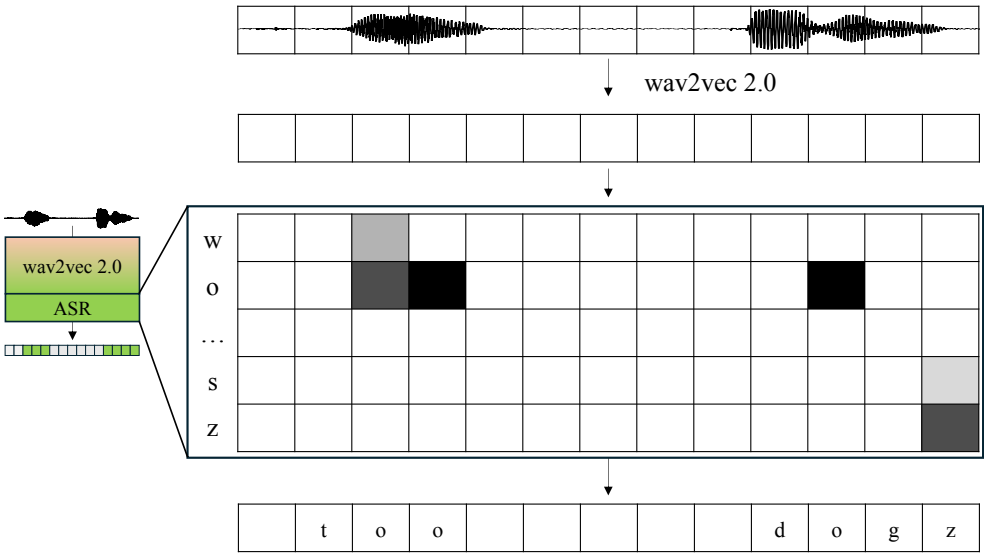


Figure 2.7: Fine-tuning a wav2vec 2.0 model for the automatic speech recognition task

[2] This 'model' may be as simple as a linear layer or as complex as a pre-trained decoder-only text transformer model.
[3] Typically using a loss such as Connectionist Temporal Classification (CTC: Graves et al., 2006).
[4] Though note however the wav2vec 2.0 encoder is typically frozen for the first 10k steps when starting with a randomly-initialised linear layer (Baevski et al., 2020, §4.3)

As seen in this transcription *too dogz*, the generated transcriptions from a wav2vec 2.0 model fine-tuned for ASR with only limited transcriptions may be rather error prone, as the system has seen very few pairs of sound and character sequences to learn from. For machine-assisted transcription to be viable (and save human effort), it is desirable for the machine-generated transcriptions to contain as few errors as possible — a transcription that is entirely incorrect may even take longer to correct than manual transcription. In Chapters 4 and 5, we examine how transcription errors can be reduced using assistive processes.

First, an 'unassisted' decoding process that is most commonly used is in the ASR literature is known as greedy decoding. As illustrated in Figure 2.7, part of the ASR process involves determining a single character for each output time step based on a probability distribution over the character set at each time step.[5] For example, at the third time step, while both *w* (light grey) and *o* (dark grey) are noted as probable at this time step, the latter has been predicted as being more likely. A similar split is found between, *s* and *z* in the last time step. In greedy decoding, the most probable character is selected as the prediction for each time step — yielding the prediction *too dogz*.

The decoding process may be assisted in two ways when text data is available from sources such as books or the internet. Using these sources, we can derive a lexicon which contains the various unique words that are possible in the language or additionally train a language model (LM) which observes which word sequences are more probable than others. Thus, a lexicon- and/or LM-assisted decoding process can limit the ASR system output to only possible words, e.g. *too dogz* → *too dogs* as well as bias the output to more probable word sequences, e.g. *too dogs* → *two dogs*. Of course, this assistive process requires that there be external text data and, importantly, the sentences be similar to those in the audio to be transcribed to avoid erroneous miscorrections. Thus, given enough transcribed speech for ASR fine-tuning, greedy decoding as good as LM-assisted decoding (Synnaeve et al., 2019).

2.3 Related work

2.3.1 Spoken term detection and open vocabulary keyword spotting

As mentioned, the detection stage in spoken term detection has conventionally been implemented using some form of Dynamic Time Warping (DTW) based algorithm.[6] More recently, there have been both neural and end-to-end approaches proposed. For example, given that this detection stage lends itself to being reformulated as a computer vision problem (i.e. detect presence of a diagonal in a 2D matrix), Ram et al. (2018) proposes replacing the DTW-based model with one based on a Convolutional Neural Network (CNN), adopting based on the computer vision literature a VGG-like model (Simonyan & Zisserman, 2015). In later work, Ram et al. (2020) propose a fully end-to-end approach without the need for a separate feature extraction stage.

[5] In some cases, it may be a set of sub-word units, see Ahmed (2023, §3.2) and works cited therein. This, however, typically assumes there is text data on which a good sub-word model can be trained and sufficient transcriptions for fine-tuning to learn mappings to a larger vocabulary size (e.g. 256 sub-word units vs. 27 characters).

[6] For a comprehensive book-length survey of the field, see Mary and Deekshitha (2018).

What is worth noting, however, is that, at least in this later study (Ram et al., 2020), the conventional two-stage system with DTW-based classification was the best performing system compared to the CNN and end-to-end ones for lower resource languages in the standard SWS2013 benchmark dataset for spoken term detection. In other words, the conventional DTW-based approach remains highly competitive when there is insufficient task-specific data for training the classifier model. To the best of my knowledge, SWS2013 and its successor QUESST2014 (which has fewer low-resource languages), remain the most current spoken term detection benchmarks.[7]

An alternative, promising starting point that could be drawn upon in future work is the keyword spotting literature, which has seen a recent interest in more open query sets. Conventionally speaking, spoken term detection is defined as a task with an open query set while its counter-part keyword search or keyword spotting is one with a limited, closed query set. Nevertheless, the open vocabulary use case is often anticipated and system evaluations may even report out-of-vocabulary performance and performance after few-shot learning on new keywords (e.g. Jung et al., 2023; Kao et al., 2023; Mazumder et al., 2021). One particularly interesting work in this area is Zhu et al. (2023), which both proposes a method for open vocabulary keyword search and releases an associated dataset with more than 115 languages, including 44 based on language documentation projects sourced from DoReCo.[8]

2.3.2 English HuBERT, WavLM, data2vec (2.0) and multilingual w2v-BERT 2.0

Since the development of wav2vec 2.0 framework by Baevski et al. (2020), there have been numerous proposals for further improving the speech pre-training process, the most directly related of which I summarise here. As introduced above, the wav2vec 2.0 pre-training process involves simultaneously learning a set of discrete units and learning to perform a masked prediction task using those units. By contrast, in HuBERT (Hsu, Bolte et al., 2021), pre-training process alternates between an acoustic unit discovery stage and the masked prediction stage, whereby acoustic units are iteratively derived from a middle transformer layer of the model trained in the previous iteration (or MFCCs for the very first iteration). Further modifying the HuBERT pre-training process, WavLM (Chen et al., 2022) proposes data augmentation along with an additional denoising objective in order to optimise representations for tasks beyond ASR (e.g. speaker verification and identification). The focus of data2vec and its successor data2vec 2.0 (Baevski et al., 2022, 2023) is to derive an efficient pre-training process for multiple modalities: vision, speech, and text. In contrast to alternating between two stages as in HuBERT/WavLM, the pre-training process leverages a teacher model that is given the whole, unmasked input and whose weights track an exponential moving average of a student model that is given a masked input and is tasked with predicting the representations of the teacher model. Compared to the 150h of compute time on 64 A100 GPUs required for pre-training the wav2vec 2.0 model, the data2vec 2.0 model for speech required only 76.7h on the same hardware while also performing better when subsequently fine-tuned for English ASR. All these models from HuBERT to data2vec 2.0, however, are pre-trained only on English.

[7] The annual MediaEval challenge which developed SWS2013 and QUESST2014 has not since featured spoken term detection again as a shared task, see https://multimediaeval.github.io/.

[8] https://doreco.info/

As mentioned, the last multilingual variant of wav2vec 2.0 is MMS, pre-trained on 491k hours of speech from 1,406 languages (Pratap et al., 2023). At the time of writing in March 2024, the latest openly available, multilingual model is w2v-BERT 2.0,[9] pre-trained on 4.5M hours released in late 2023 as part of a suite of models by Barrault et al. (2023, see §3.2.1). The notable architecture changes in this model is the use of a joint contrastive loss (similar to wav2vec 2.0, without the diversity loss) and masked prediction loss (similar to HuBERT) though without alternating between two stages (unlike HuBERT), and the use of conventional filterbank features instead of a convolutional feature extractor.[10] Additionally, in place of a 24-layer transformer network is an equivalent one composed of conformer layers (Gulati et al., 2020), which leverage convolution neural networks to better model local features and complement transformer layers that model the global context.

[9] https://huggingface.co/facebook/w2v-bert-2.0

[10] See Parcollet et al. (2023) for wav2vec 2.0 pre-training experiments with different feature extraction methods showing minimal degradation in downstream ASR performance, especially considering the compute cost.

Chapter 3

Leveraging pre-trained representations to improve spoken term detection

As recording spoken language is much easier than transcribing the recorded speech, language documentation efforts often yield a substantial amount of untranscribed speech, which is difficult to index and search. Yet how easily recorded speech can be searched and retrieved has a direct impact on how effectively such resources may be used by all interested parties — from linguists, to language teachers, to community members. As introduced in the first chapter, one approach to improve access could be through query-by-example spoken term detection (QbE-STD), which as illustrated in Figure 3.1, enables the use of voice queries to perform searches on untranscribed speech corpora. Recent work trialling QbE-STD for language documentation corpora have found reduced performance when the speaker of the query was different to those in the corpora. In this chapter, I examine whether leveraging pre-trained representations can help improve QbE-STD in these scenarios.

3.1 Background

3.1.1 What is query-by-example spoken term detection?

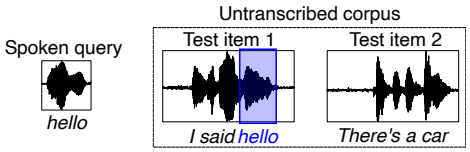


Figure 3.1: An illustration of Query-by-Example Spoken Term Detection (QbE-STD)

In querying a corpus of speech data, we are often interested in finding all instances of a particular spoken form or similar sounding utterances that may constitute minimal pairs — but how might we accomplish such

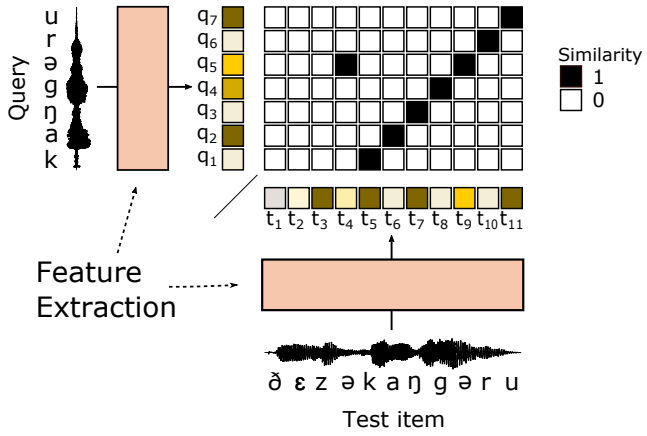


Figure 3.2: An illustration of the feature extraction stage for Query-by-Example Spoken Term Detection (QbE-STD)

searches in untranscribed recordings? Query-by-example spoken term detection (QbE-STD), which, as illustrated in Figure 3.1, is a long-standing speech information retrieval task of finding all regions within a corpus of audio recordings where a spoken query term occurs (Fiscus et al., 2007; Myers et al., 1980; Rodriguez-Fuentes et al., 2014; Rohlicek, 1995). In standard terminology, each audio recording in which the spoken query is searched for is considered a 'test item'. A conventional QbE-STD system consists of two broad stages. In the first stage, information or 'features' are extracted from both the query and test items. The second stage involves calculating a detection score of how likely a query occurs in a test item for each pair of query and test item, typically using a Dynamic Time Warping (DTW) based template matching.

In this chapter, we focus on the feature extraction stage, whose goal is to extract information from the speech signals of the query and test items such that the similarities in their speech content can be compared. As illustrated in Figure 3.2 using a query *kangaroo* [kaŋgəru] and test item *there's a kangaroo* [ðɛzəkaŋgəru], the ideal feature extraction process would yield a representation of each that facilitates the correct detection in the second stage. Roughly summarised, the task of the second stage is to use the representations to look for whether there is a region of high similarity between the query and test item, as illustrated by the diagonal in Figure 3.2.

As is the case for many tasks involving speech, a noise-robust and speaker-invariant feature extraction method is a key component of a performant QbE-STD system. Indeed, the experiments in Le Ferrand et al. (2020) showed that while QbE-STD seemed promising for facilitating searches on speech from language documentation efforts, considerably lower retrieval performance was observed when the speaker of the query and that of the test item were different. In this chapter, we are interested in whether representations extracted using large-scale pre-trained models are speaker-independent and noise-robust such that they help improve QbE-STD performance for this use case.

3.1.2 Related work

As mentioned, a spoken term detection system is conventionally composed of two main stages: feature extraction and detection. One motivation for focusing on feature extraction is that recent work has shown that for languages without task-specific data to train a neural network based detection system, the conventional, non-neural Dynamic Time Warping (DTW) approach remains extremely competitive. For example, recent work has examined using a neural detection system based on a convolutional neural network (CNN: Ram et al., 2018) or forgoing conventional stages entirely by using an end-to-end system (Ram et al., 2020). However, not all languages benefit equally from these advances. While the proposed neural systems did outperform the conventional DTW system for larger European languages from the SWS2013 QbE-STD benchmark dataset, the highest performing system for all of the smaller African languages was the conventional DTW system.

As mentioned, obtaining speaker-independent and noise-robust representations is a critical part of developing a spoken term detection system. Conventionally, this goal was achieved by training a 'bottleneck' feature extractor for phone classification. That is, the network is designed with an intentional bottleneck which restricts information flow (using a hidden layer that has a lower dimensionality compared to the surrounding layers) and tasked with identifying phones in speech signals. This design encourages the model to discard information that is irrelevant to phone classification (e.g. speaker-specific information). However, training a bottleneck feature extractor typically requires speech labelled at the phone level, which is costly to generate even for major languages, restricting their availability to all but the largest of languages. Given that self-supervised models such as wav2vec 2.0 do not require human labelled data for training, we are interested in discovering whether representations learned in this way are useful for the spoken term detection task, especially for languages for which bottleneck feature extractors are not readily available.

Results from Bartelds et al. (2022) suggest that representations extracted from a transformer layer of wav2vec 2.0 (w2v2) models may indeed be useful for audio-on-audio comparisons such as spoken term detection. Experiments in Bartelds et al. (2022) investigated which of the 24 w2v2 Transformer layers may be best suited for automatic pronunciation scoring of non-native English speech. Similar to QbE-STD, the task of pronunciation scoring is a 2-stage process: features are extracted from native and non-native speech samples of the same read text, and then a DTW-based distance is calculated (lower distance indicates closer pronunciations). Results from Bartelds et al. (2022) showed that distances between native and non-native speech samples had a strong negative correlation with native speaker ratings of the same samples (r = -0.85; lower distances correspond to higher ratings) when computed using representations extracted from the 10th Transformer layer. In other words, the representations of the middle layers of the w2v2 Transformer network appear to be well-suited for comparing phonetic characteristics of speech samples while being robust to recording conditions and speaker characteristics.

3.2 Method

As mentioned, previous experiments in Le Ferrand et al. (2020) trialling QbE-STD with language documentation data showed that occurrences of a query were less likely to be detected when the query and test item being compared were spoken by different speakers. Our own pilot experiments with Mel-frequency cepstral coefficients (MFCC) and bottleneck features (BNF), which we report here as baselines, had also shown that even when the speaker was the same, acoustic differences arising from recording equipment and/or environment differences can result in similar drops in performance. Motivated by these issues, we examined whether performance improved with features extracted from various stages of the w2v2 English and multilingual models for QbE-STD using datasets in which the speakers and/or recording conditions between the query and test items were the same, overlapping, or different. We examined both the English and multilingual w2v2 models as earlier work had shown that for keyword spotting (i.e. QbE-STD with a closed query set) in West African languages the English w2v model performed on par with a w2v model trained on 10 West African languages (Doumbouya et al., 2021).

3.2.1 Datasets

We curated data from English, Gronings, and 7 Australian Aboriginal languages. Gronings and all of the Australian languages are currently classified as vulnerable or endangered (Eberhard et al., 2020). For these languages, the audio were sourced from various language documentation projects (e.g., audio for dictionaries, audio book of short stories) and thus constitute in-domain test data for the QbE-STD systems evaluated.

We gathered for each language queries from 1-5 speakers consisting of words and multi-word expressions

Dataset	Language	Number of queries (Speakers)	Number of test items (Speakers; Duration)	Query vs. Test items	
				Speakers	Recordings
gbb-pd	Kaytetye	397 (1F)	397 (1F; 16 mins)	Same	Same
wrm-pd	Warumungu	383 (1F)	383 (1F; 20 mins)	Same	Same
wrl-mb	Warlmanpa	23 (1F)	162 (1F; 11 mins)	Same	Different
gup-wat	Kunwinjku	50 (5M)	725 (5M, 2F; 32 mins)	Overlap	Overlap
gbb-lg	Kaytetye	189 (1F)	809 (8F, 4M; 45 mins)	Overlap	Different
pjt-sw01	Pitjantjatjara	30 (1F)	320 (5F; 38 mins)	Different	Different
mwf-jm	Murrinhpatha	37 (1F, 1M)	259 (3F, 2M; 13 mins)	Different	Different
wbp-jk	Warlpiri	24 (1F)	198 (12F; 6 mins)	Different	Different
gos-kdl	Gronings	83 (2F, 3M)	430 (2F, 2M; 23 mins)	Overlap	Overlap
eng-mav	English	100 (1M)	526 (1M; 77 mins)	Different	Different

Table 3.1: Characteristics of evaluation datasets. Parentheticals indicate speaker compositions, and total length of test audio in minutes. Same, overlap, or different indicate whether the queries and test items shared speakers and/or recording conditions.

and test items consisting of longer utterances from 1-12 speakers. Each dataset varied according to whether the query and test items shared speaker(s) and/or recording conditions, as indicated in Table 3.1. For example, for gbb-lg, the 189 queries were sourced from audio produced in a recording studio by a single female speaker AR. The 809 test items were sourced from various field recordings from 12 speakers (8F, 4M: one of whom was AR). Thus, for gbb-lg, the queries and test items overlapped in speakers but differed in recording conditions.

3.2.2 Pre-trained models

In our feature extraction procedure, we use two w2v2-based models, namely the English monolingual model described in Baevski et al. (2020) and the multilingual 'XLSR53' model described in Conneau et al. (2021). The English monolingual model is pre-trained on the unlabelled Librispeech dataset (LS960), which contains 960 hours of clean and noisy English speech from audio books. The multilingual model is pre-trained on 56,000 hours of data from 53 different languages (amount of data varies per language, though 44,000 hours of English from the Multilingual Librispeech dataset constitutes the majority), and includes African, Asian, and European languages obtained from audio books, conversational telephone speech, and read speech. The architecture of both models is the same, except for the quantisation of the encoder representations. XLSR53 learns a single set of quantised speech representations on the basis of the encoder output. This set was subsequently shared across languages, which was shown to be effective for learning self-supervised multilingual speech representations for ASR.

3.2.3 Procedure

To create ground truth labels, we first paired each query's text transcription with that of each test item within a given dataset. Then for each pair, we used regular expressions to determine whether the query occurred in the test item. All audio data from various sources were standardised to mono 16-bit PCM at 16 kHz (sample rate required by w2v2). For MFCC and bottleneck features (BNF), the audio was further downsampled to 8 kHz at extraction time to meet the requirements of the respective feature extractors.

For baseline comparisons, we extracted MFCC and BNF representations using the Shennong library,[1] which provides a Python interface to common feature extraction routines. For MFCC, we used the Kaldi feature extractor which returns for each time frame 13 MFCC features with their first and second derivatives (39 features total). For BNF, we used the BUT/Phonexia feature extractor which returns for each time frame 80 activation values from a bottleneck layer originally trained for phone classification on the 17 languages of the IARPA Babel dataset (Silnova et al., 2018). For w2v2 features (both from the English monolingual model and XLSR53 multilingual model), we adapted the feature extraction code from Bartelds et al. (2022),[2] which extracts outputs from the Encoder CNN (E), the Quantiser module (Q), or any one of the 24 layers of the Transformer network (T01–T24). Thus, in total we extracted features using 54 different methods for each of the 10 datasets.

[1] https://docs.cognitive-ml.fr/shennong/
[2] https://github.com/Bartelds/neural-acoustic-distance

As our primary interest is in comparing different feature extraction methods, we implemented a DTW-based detection stage based on a state-of-the-art QbE-STD system (Rodriguez-Fuentes et al., 2014). Given two feature matrices, of a query Q and of a test item T, we calculated standardised Euclidean distances between each time frame in Q and T. To make distances between different pairs of queries and test items comparable, these distances were range normalised to [0, 1]. A window the size of the query was moved along the normalised distance matrix and a DTW-based distance between the query and subpart of the test item was calculated at each step. The score for how likely a query occurs in a test item was then calculated as 1 minus the minimum distance.

We used the Maximum Term Weighted Value (MTWV) to evaluate each of our QbE-STD systems (e.g. MFCC+DTW, w2v2-E+DTW, etc.). The MTWV ranges from 0, indicating a system that simply returns nothing, to 1, indicating a perfect system detecting all relevant instances with no false positives. Using the NIST STDEval tool,[3] we calculated MTWVs with the suggested NIST costs (false positive: 1, false negative: 10) and an empirical prior of 0.0278 (true positive rate averaged across all 10 datasets). With these costs, a MTWV of 0.48 indicates a system that correctly detects 48% of all queries searched,[4] while producing at most 10 false positives for each true positive correctly retrieved (White et al., 2015).

For significance testing, we computed the MTWV for each query in a given dataset and performed a one-sided paired *t*-test to examine whether the MTWVs obtained for the queries using one feature extraction method were significantly greater than those obtained using another method.

With the exception of the raw audio and transcriptions from the Australian language documentation projects for which we do not have permissions to release openly, we have made all pilot experiments results, experiment code and analyses available on a GitHub repository, and placed the Gronings data and all experimental artefacts in Zenodo archives.[5] The English data can be obtained from the Mavir corpus (Sandoval & Llanos, 2012).[6]

3.3 Results and discussion

Figure 3.3 displays spoken term detection performance as measured by MTWV (y-axis; higher is better), and how this performance varies across the 10 datasets (indicated by shapes) and various feature extraction methods (x-axis: MFCC, BNF), including those extracted from various layers (E, Q, T01, ..., T24) from two wav2vec 2.0 models (LS960, trained only on English; XLSR53, trained on 53 languages). As expected, dataset characteristics had a clear effect on QbE-STD performance. For our baselines, retrieval performance using MFCC was highest on the datasets having the same speaker and recordings between the queries and test items (wrm-pd: 0.83; gbb-pd: 0.71), lower on those with overlapping speakers (e.g. wrl-mb: 0.48) or recordings (e.g. gup-wat: 0.33), and lowest when both were entirely different (e.g. wbp-jk: 0.18, mwf-jm: 0.14). The

[3] https://www.nist.gov/itl/iad/mig/tools

[4] To the best of my knowledge, there have not been user studies examining the relationship between MTWV and usability of a spoken term detection system (e.g. analogous to those in automatic speech recognition and the word error rate metric).

[5] All linked on: https://github.com/fauxneticien/qbe-std_feats_eval

[6] http://www.lllf.uam.es/ING/CorpusMavir.html

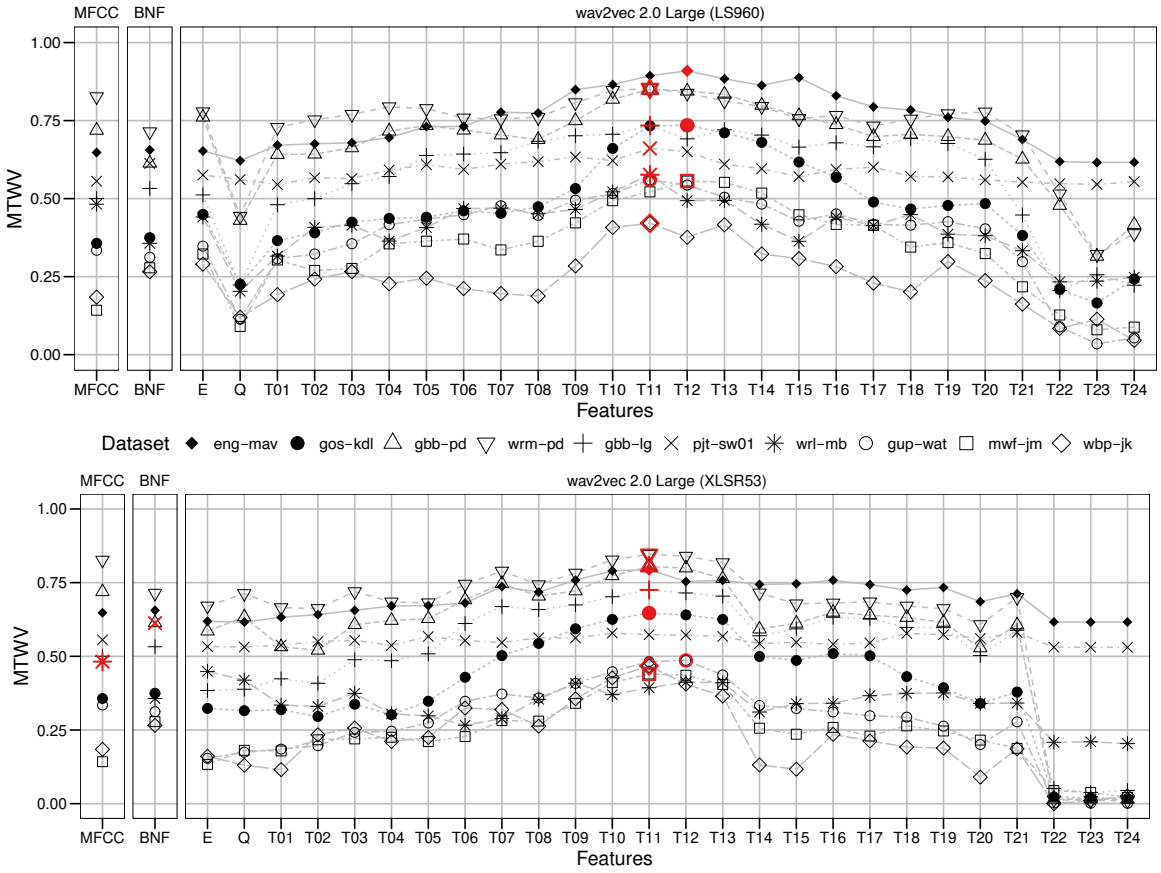


Figure 3.3: Maximum Term Weighted Values achieved on a QbE-STD task using various speech representations: baselines (MFCC; bottleneck features: BNF) and from layers (E; Q; T01-24) of two wav2vec 2.0 models (English monolingual: LS960; multilingual: XLSR53). Shapes indicate one of 10 datasets; red-coloured/largest shape indicates highest score achieved on dataset.

same effects were also seen using BNF, which improved on different-speaker/recordings datasets (e.g. wbp-jk: 0.27), though at the cost of worsening on those with same-speaker/recordings (e.g. wrm-pd: 0.71).

Using outputs from a middle Transformer layer of a w2v2 model resulted in increased performance on nearly all our datasets. For 9 of 10 datasets, the MTWVs obtained using outputs of the Transformer layer 11 from the English monolingual model were significantly greater than those obtained using BNF, as indicated by the t-tests in Table 3.2 (b. LS960-T11 $>$ a. BNF). Similarly, the MTWVs obtained using Transformer layer 11 of the multilingual model were significantly greater than those using BNF for 8 of 10 datasets (Table 3.2: c. XLSR53-T11 $>$ a. BNF). Overall, the best performing representations were those from the Transformer layer 11 of the English w2v2 model, which also significantly outperformed those from the equivalent layer of the multilingual w2v2 model for 4 of the 8 Australian language datasets (Table 3.2: b. LS960-T11

Dataset (N queries)	MTWV (averaged over queries)			One-sided paired t-test (t-value, p-value)		
	a. BNF	b. LS960-T11	c. XLSR53-T11	H_1: b. > a.	H_1: c. > a.	H_1: b. > c.
gbb-pd (397)	0.612	0.849	0.803	(14.1, .000)	(11.1, .000)	(3.8, .000)
wrm-pd (383)	0.713	0.853	0.843	(8.7, .000)	(8.2, .000)	(0.7, .236)
wrl-mb (23)	0.352	0.573	0.379	(3.0, .003)	(0.3, .372)	(3.4, .001)
gup-wat (50)	0.313	0.561	0.483	(6.6, .000)	(3.9, .000)	(2.7, .005)
gbb-lg (189)	0.533	0.733	0.719	(6.7, .000)	(6.6, .000)	(0.5, .299)
pjt-sw01 (30)	0.607	0.660	0.572	(1.4, .080)	(-0.9, .082)	(2.2, .020)
mwf-jm (37)	0.279	0.515	0.435	(4.8, .000)	(2.4, .010)	(1.3, .102)
wbp-jk (24)	0.267	0.422	0.468	(1.9, .037)	(2.6, .009)	(-0.6, .732)
gos-kdl (83)	0.372	0.728	0.647	(9.8, .000)	(8.2, .000)	(2.9, .002)
eng-mav (100)	0.656	0.894	0.794	(11.1, .000)	(5.7, .000)	(3.9, .000)

Table 3.2: Results of one-sided t-tests for the alternative hypothesis (H_1) that the Maximum Term Weighted Value (MTWV) obtained for each query in a given dataset is greater using different speech representations: baseline bottleneck features (BNF), and Transformer layer 11 (T11) of two wav2vec 2.0 models (English monolingual: LS960; multilingual: XLSR53).

> c. XLSR53-T11). On the two most challenging Australian language datasets with different speakers and recordings, performance increased 56-86% using T11 features of the English model (wbp-jk: 0.42, mwf-jm: 0.52) over the best achievable baseline scores using BNF (wbp-jk: 0.27, mwf-jm: 0.28); n.b. the English and multilingual models did not differ significantly on these two datasets. These results demonstrate the effectiveness of contextualised representations learned by the Transformer network for comparing the phonetic characteristics of speech samples while being robust to recording conditions and speaker characteristics.

To better understand the representations of the Transformer layer 11 of the English w2v2 model, we undertook an error analysis of the erroneous retrievals by the QbE-STD system on the Australian languages when using these features.We focused on 'unretrievable' queries, defined as those for which no true matches were present in the top 5 results returned for that query, and examined the differences in phonetic transcriptions between the query and its top match. This analysis revealed that for the majority of these queries (89/119 = 75%), there were 4 or fewer segmental differences between the transcriptions of the query and its top match, and that, notably, the query and its top match typically shared a manner template. For example, for the Kaytetye query [aɳanpə] 'medicinal sap', its top match was the word [anənkə] 'to sit', both of which share the VNVNTV template (where V represents a vowel, N a nasal, and T a plosive). In other words, the representations of the w2v2 English model do not appear to be sufficiently fine-grained to differentiate between segments differing primarily in place of articulation, especially for those that do not occur contrastively in English (e.g. retroflex nasal [ɳ] vs. alveolar nasal [n]).[7]

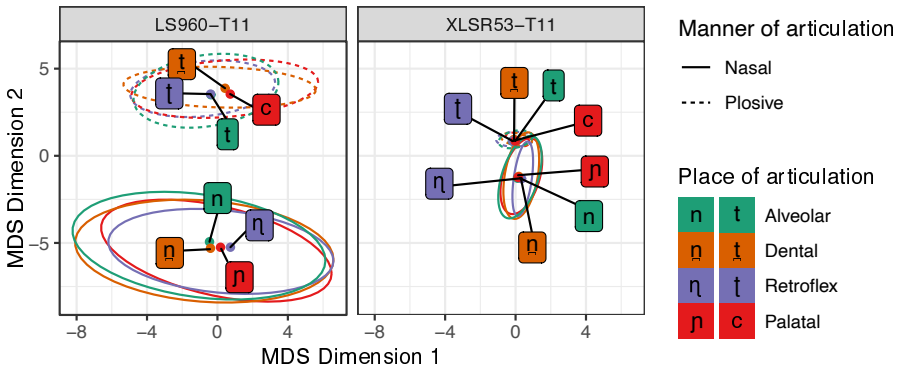


Figure 3.4: Multidimensional scaling (MDS) visualisations of features extracted from Kaytetye consonants (data from Harvey et al., 2015) using Transformer layer 11 (T11) of wav2vec 2.0 (English monolingual: LS960; multilingual: XLSR53), averaged over the duration of the consonant. Ellipses represent 95% confidence intervals and text labels represent means.

3.3.1 Exploring wav2vec 2.0 features

Based on the findings of our error analysis, we examined whether differences in the way phonetic information is encoded by the w2v2 English and the multilingual models could explain differences in their QbE-STD performance. We used the Transformer layer 11 of the English and multilingual models to extract features from Kaytetye consonants (from audio stimuli collected by Harvey et al., 2015). Figure 3.4 displays the dissimilarities between each of the Kaytetye consonant types in the 1024-dimensional space of the two w2v2 models (averaged across the duration of the consonant and reduced to 2 dimensions through multidimensional scaling). For considerations of space and visual clarity, we focus on the nasals and plosives.

Despite being given no phone labels at training time, the w2v2 models learn to encode broad manner of articulation classes within its representations, as evidenced by ellipses forming two clusters for nasals (solid lines) and plosives (dashed lines) in Figure 3.4. In line with findings from our qualitative error analysis, features from the w2v2 English model under-differentiate place of articulation contrasts, as evidenced by the overlapping ellipses within each of the nasal and plosive manners of articulation in Figure 3.4. Compared to the English model, the representations of the Kaytetye consonants by the multilingual model overlap to a larger degree. This result suggests that representations extracted from a model trained on a single language or a set of phonologically similar languages may be more beneficial for QbE-STD than a large multilingual model such as XLSR53 trained on a diverse set of 53 languages — especially when supervised fine-tuning in the target languages is not a readily available option.

[7]For an experimental investigation of comparing how native and non-native speech perception patterns in humans compare with in- and out-of-domain languages for self-supervised speech models see Millet and Dunbar (2022) and Millet et al. (2021)

3.4 Conclusion

In this chapter, we compared the performance of 54 feature extraction methods for QbE-STD using data from 9 languages, eight of which are vulnerable or endangered. Our results showed that the middle layers of the wav2vec 2.0 Transformer network provide a noise-robust and speaker-invariant representation of speech sounds that can be leveraged for QbE-STD with low resource languages, performing significantly better than state-of-the-art approaches using bottleneck features. An error analysis and a preliminary investigation of the information encoded in the representations learned by the wav2vec 2.0 Transformer network suggested that the representations extracted from the English model may provide better features for QbE-STD in unrelated languages than the multilingual model in cases where important phonological distinctions in those languages are not as distinctively encoded by the multilingual model. Given the alternative is browsing untranscribed audio in near real-time, QbE-STD does help provide better access via enabling voice searches. However, the inherent difficulty of an audio-on-audio the matching task may limit its capabilities, especially on noisy field recordings in archival corpora.

Chapter 4

Leveraging pre-trained models to derive a metalanguage proxy index

Many archival recordings of speech from endangered languages remain unannotated and inaccessible to community members and language learning programs. One bottleneck is the time-intensive nature of annotation. An even narrower bottleneck occurs for recordings with access constraints, such as language that must be vetted or filtered by authorised community members before annotation can begin. In this chapter, I develop and validate a privacy-preserving workflow to widen both bottlenecks for recordings where speech in the target language is intermixed with a more widely-used language such as English for meta-linguistic commentary and questions (e.g. *What is the word for 'tree'?*). Given such a corpus, I propose automatically isolating and transcribing the metalanguage content, which an authorised person can quickly scan to triage recordings that can be annotated by people with lower levels of access. These transcriptions also provide approximate locations where certain target language words and phrases are being discussed, providing a searchable index of the corpus, streamlining the production of language learning materials. In this chapter, I report on the development of this pipeline and its use in an ongoing documentation project to process 136 hours archival audio containing a mix of English and Muruwari, an Australian Aboriginal language.

4.1 Background

In speech recorded for language documentation, it is common to find not only the target language that is being documented but also a language of wider communication, such as English. This is especially so in early-stage fieldwork when the elicitation may centre around basic words and phrases from a standard word list Swadesh (e.g. the Swadesh List: 1955). In these mixed-language recordings, utterances in the language of wider communication are largely meta-linguistic questions and commentary (e.g. *What is the word for 'tree'?*, *This word means 'soft'*), which appear inter-mixed with the utterances of interest in the target language. In this chapter, I propose a workflow to help process hundreds of hours of unannotated speech of this genre.

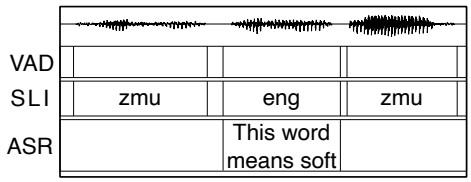


Figure 4.1: Deriving transcriptions of English in mixed-language speech using voice activity detection (VAD) and spoken language identification (SLI) to identify speech regions and the language spoken (zmu: Muruwari or eng: English) and automatic speech recognition (ASR) to transcribe English speech.

I describe a use case where the language of wider communication is English (ISO 639-3: eng), and the documented language is Muruwari (ISO 639-3: zmu), an Aboriginal language traditionally spoken in north western New South Wales, Australia. As illustrated in Figure 4.1, I propose using voice activity detection (VAD) to detect speech regions, then spoken language identification (SLI) to distinguish between Muruwari and English regions, and then automatic speech recognition (ASR) to transcribe the English. The uncorrected transcriptions offer a rough but workable estimate of the contents in a given recording.

This workflow is being used to help process 136 hours of predominantly single-speaker recordings made in the 1970s by the last first language speaker of Muruwari, James 'Jimmie' Barker (1900-1972). The generated transcriptions can be used by the data custodian and Muruwari elder, Roy Barker, to triage the recordings and make initial decisions on which recordings can be listened to by people with lower levels of access who can then correct the transcriptions. The corrected transcriptions provide approximate locations where certain Muruwari words and phrases are being discussed, providing an index of the corpus from which language learning materials can be produced. In this way, this machine-assisted workflow is able to support ongoing language revival initiatives through a strategic deployment of machine and human efforts in a manner that adheres to the level of privacy required.

4.1.1 Muruwari documentation project

Muruwari is an Aboriginal language traditionally spoken in north western New South Wales, Australia and belongs to the Pama-Nyungan family of Australian languages (Oates, 1988). Oates (1988), which comprises the largest extant single work on Muruwari, describes it as a relative isolate compared to the neighbouring Pama-Nyungan languages, Yuwaaliyaay, Yuwaalaraay, Barranbinya, Ngiyampaa (Ngemba), Guwamu and Badjiri.

James 'Jimmie' Barker (1900–1972), the last first language (L1) speaker of Muruwari, produced in the early 1970s a total of 136 hours of reel-to-reel tape recordings consisting of a mix of Muruwari and meta-linguistic commentary on the Muruwari language in English. The now digitised recordings are held at the Australian Institute of Aboriginal and Torres Strait Islander Studies and access to these materials depend on permission from the custodian and Muruwari elder, Roy Barker (RB: grandson of Jimmie Barker).

To date, RB has manually auditioned approximately 40 of the 136 hours over the course of 4 years to determine regions of speech appropriate for general access and those requiring restricted access (e.g. for only the Muruwari community, or only the Barker family). At this rate of roughly 10 hours per year, the remaining 96 hours may require nearly an additional decade of manual review by RB. Parallel to the review of the remaining recordings, a subset of the recordings that have already been cleared by RB is being used to search for excerpts that may be useful for learning materials and those that can inform the development of a standardised orthography for Muruwari, as desired by the Muruwari community. To support these efforts and in collaboration with RB, we investigated how to combine voice activity detection (VAD), spoken language identification (SLI), and English automatic speech recognition (ASR) in order to allow for the review process and excerpt searches to be done more strategically and efficiently.

4.1.2 Related work and research questions

There has been growing interest in leveraging speech processing tools to assist in language documentation workflows, including the formulation of shared tasks (e.g. Levow et al., 2021; Salesky et al., 2021).[1] Aimed at making unannotated fieldwork recordings more accessible, Levow et al. (2017) proposed a family of shared tasks, dubbed the "Grandma's Hatbox", which include SLI and ASR. In our work, we additionally leverage VAD to make the system fully automatable and, to derive a rough index of the corpus, we transcribe all speech regions detected as English (in the shared task formulation, ASR was intended to transcribe only the metadata preamble in the recordings).

The performance of speech processing systems can be poor when there are mismatches between the speech on which they were trained and that on which they are deployed. Commenting on such poor deployment-time performance of SLI systems, Salesky et al. (2021) concluded that what is necessary for real-world usage are methods for system adaptation with a few examples from the target speakers/domains. Accordingly, we sought to answer the following questions: 1) How many utterances of English and Muruwari are needed to adapt an off-the-shelf SLI system? 2) Is it possible to make use of such a system without compute-intensive adaptation methods requiring a graphics processing unit (GPU)?

Regarding this latter question, we were inspired by a recent probing study on various speech representations showing that logistic regression classifiers performed on-par with shallow neural networks for two-way classification of speech, e.g. distinguishing between vowels and non-vowels (Ma et al., 2021). Hence, we examined through our SLI experiments whether using a logistic regression classifier suffices for the two-way classification of the speech data, i.e. distinguishing between English and Muruwari.

Turning now to ASR, the typical use case in language documentation work has been to develop ASR systems to help transcribe the target language (e.g. Adams et al., 2018; Prud'hommeaux et al., 2021; Shi et al., 2021). By contrast, our use of ASR more closely aligns with recent work exploring techniques such as spoken term detection to help locate utterances of interest in untranscribed speech corpora in the target

[1] Aimed to help drive system development, shared tasks are competitions in which teams of researchers submit competing systems to solve a pre-defined challenge.

languages (Le Ferrand et al., 2020, 2021; San et al., 2021). In this work, however, we take advantage of the mixed-language speech in the corpus, and leverage SLI and ASR to transcribe the English speech as a way to derive a rough index.

We opted to use the Robust wav2vec 2.0 model (Hsu, Sriram et al., 2021) to reduce the mismatch in audio quality between the training and the deployment data (i.e. noisy archival recordings). This model is pre-trained not only on LibriSpeech (960 hours: Panayotov et al., 2015) and CommonVoice English (700 hours: Ardila et al., 2020), but also on noisy telephone-quality speech corpora (Fisher, 2k hours: Cieri et al., 2004 and Switchboard, 300 hours: Godfrey et al., 1992), and also fine-tuned on 300 hours of transcribed speech from Switchboard. With our ASR experiments, we sought to answer the following questions: 1) What amount of transcribed speech is sufficient to reliably achieve better than off-the-shelf performance? 2) Using the same amount of transcribed speech, to what extent can ASR system performance be further increased when supplemented with a language model trained on external texts?

4.2 Dataset: the Jimmie Barker recordings

In this section, we describe the data curation and preparation procedures common to subsequent experiment sections (Experiment 1: SLI; Experiment 2: ASR; Experiment 3: timed annotations). To gather training and evaluation data for the two speech processing tasks, we obtained 6 archival recordings of Jimmie Barker's speech cleared by RB. For each recording, we used the off-the-shelf Robust wav2vec 2.0 (Hsu, Sriram et al., 2021),[2] to simply transcribe all speech regions detected by the Silero VAD system,[3] and generated an .eaf file for ELAN.[4] Using ELAN, three annotators (2 recordings per annotator) then erased the spurious text for the Muruwari utterances (i.e. for SLI, we simply used blank annotations to denote Muruwari regions, given the orthography is still in development) and manually corrected the English transcriptions for ASR (i.e. for SLI, any non-blank region with text was considered English). While the machine-generated annotations for the training and evaluation data were human-corrected, we have yet to establish inter-annotator agreement or conduct error analyses.

When correcting the English transcriptions, speech was transcribed verbatim with no punctuation except for apostrophes, i.e. including false starts (e.g. *we we don't say*) and hesitations (e.g. *and uh it means steal*). To facilitate searches, transcriptions were made in lower-case with the exception of proper nouns (e.g. *uh the Ngiyaamba has it uh*) and words that were spelled out by Jimmie (e.g. *you've got B U at the end of a word*). For ASR training, the transcriptions were automatically converted to all upper-case to normalise the text to a 27-character vocabulary (26 upper-case letters + apostrophe) that matches vocabulary with which the wav2vec 2.0 Robust model was originally trained.

Based on the corrected annotations, we extracted the speech regions into individual 16-bit 16 kHz .wav files and all the transcriptions for the English utterances into a single tab-delimited file. A summary of the data

[2] https://huggingface.co/facebook/wav2vec2-large-robust-ft-swbd-300h
[3] https://github.com/snakers4/silero-vad
[4] https://archive.mpi.nl/tla/elan

used in this paper is given below in Table 4.1. Overall, the yielded speech content contained more English than Muruwari (78% English by duration or 66% by number of utterances), reflecting the relatively more numerous and longer nature of the meta-linguistic commentary in English compared to the Muruwari words and phrases being commented upon.

Recording ID (Running time, mins)	**Speech (mins)** eng	zmu
33-2162B (65)	23.2	2.06
31-1919A (65)	16.3	6.28
25-1581B (65)	15.5	4.75
25-1581A (65)	12.1	4.34
28-1706B (64)	7.00	2.06
25-1582A (35)	6.92	2.68
Total: 5.98 hours	81.0 mins	22.2 mins
4864 utts.	3243 utts.	1621 utts.

Table 4.1: Duration and number of utterances (utts.) of English and Muruwari speech yielded from labelling 6 archival recordings

Notably, only a third of the total running time of the recordings was found to be speech content on average, with frequent inter- and intra-phrase pauses arising from the semi-improvised linguistic self-elicitation being undertaken by Jimmie. A consequence of these pauses is that the VAD system segments Jimmie's speech into sequences of sentence fragments, e.g. *This word...*, *This word means soft...*, *And also softly*. We will return to these data characteristics in our discussion of the timed annotation tasks.

Finally, we note that having had few prior experimentally-informed estimates of the minimum amounts of data required, we chose to label for our initial implementation of this workflow this specific set of 6 recordings in accordance with other project priorities. While our deployed models are those trained on all the data, we opted to run detailed analyses on how much of the labelled data was actually necessary for adapting the SLI and ASR models to help establish estimates regarding the minimum amounts of labelled data needed to apply this workflow in other settings, and timed the annotation tasks using models trained on these minimum amounts of data.

4.3 Experiment 1: Spoken Language Identification

We are interested in finding the minimum amount of training utterances required to obtain a performant system for same-speaker SLI. As training a system with very few utterances can lead to a large variance in its performance on unseen utterances, we were particularly interested in determining the training set size at which the variance was functionally equivalent to training on all available data.

4.3.1 Method

For our SLI experiments, we first extracted speech representations from each of the 4864 English and Muruwari utterances using the SpeechBrain toolkit (Ravanelli et al., 2021), which includes a state-of-the-art SLI model trained on 107 languages of the VoxLingua107 dataset (Valk & Alumäe, 2021).[5] We then performed 5000 iterations of training and evaluating logistic regression classifiers. At each iteration, the dataset was shuffled and 20% of the data (972 utterances) was held out as the test set. The remaining 80% of data (3892 utterances) was designated as the 'All' training set and from which we sampled 5 additional subsets (1, 5, 10, 25, and 50 utterances per language). We trained separate logistic regression classifiers using each of the 6 datasets (5 subsets + All), and then measured SLI performance of each classifier on the same test set using the F1 score.[6] Finally, we also calculated the differences between the F1 scores for the classifier trained on all the training data and each of those trained on the smaller datasets (All vs. 1, All vs. 5, All vs. 10, All vs. 25, All vs. 50).

4.3.2 Results

Figure 4.2 displays the mean F1 scores for each of the training dataset sizes. The error bars represent the 95% bootstrap confidence interval (CI) for the mean obtained over 5000 iterations. Using all the training data

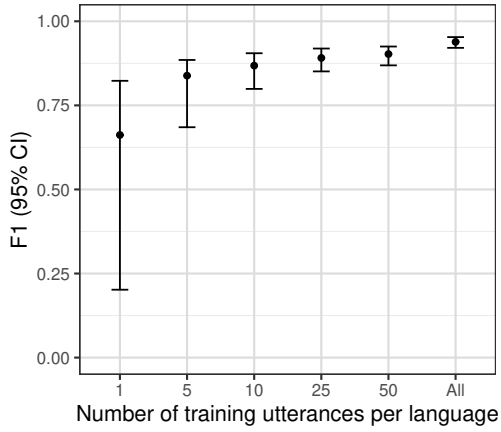


Figure 4.2: Two-way spoken language identification performance (English vs. Muruwari) using logistic regression classifiers trained on SpeechBrain SLI embeddings (Ravanelli et al., 2021) using varying dataset sizes (1, 5, 10, 25, 50 utterances per language, and All available data: 3892 utterances). Points represent mean F1 and error bars the 95% bootstrap confidences intervals over 5000 iterations.

[5] While the model was trained to identify English (dialects unspecified), we found that the included, off-the-shelf classifier could not consistently identify Jimmie's Australian English utterances, which were most frequently classified as Welsh (497/3243: 15.3%) or English (321/3243: 9.8%).

[6] Ranging between 0 (worst) and 1 (best), the F1 score is a measure of a classification system's accuracy, taking both false positives and false negatives into account.

Comparison	Difference in F1 Mean, [95% CI]: CI width
a. All vs. 1	0.28, [0.11, 0.74]: 0.63
b. All vs. 5	0.10, [0.05, 0.25]: 0.20
c. All vs. 10	0.07, [0.03, 0.14]: 0.11
d. All vs. 25	0.05, [0.02, 0.09]: 0.07
e. All vs. 50	0.04, [0.01, 0.07]: 0.06

Table 4.2: Mean difference in F1 and 95% bootstrap confidence intervals (lower and upper bounds, and width) for the difference in means for the performance on a spoken language identification task using logistic regression classifiers trained of varying dataset sizes (1, 5, 10, 25, 50 utterances per language, and All available training data: 3892 utterances)

resulted in the highest SLI performance of 0.93 [95% CI: 0.91, 0.95]. Of the smaller dataset sizes, the 50-, 25-, and 10-utterance training subsets performed similarly with mean F1 scores of 0.90 [95% CI: 0.87, 0.93], 0.89 [95% CI: 0.85, 0.92], and 0.87 [95% CI: 0.79, 0.91], respectively. The smallest two dataset sizes showed yet lower SLI performance with mean F1 scores for 5 utterances at 0.84 [95% CI: 0.69, 0.89] and 1 utterance at 0.66 [95% CI: 0.20, 0.82].

Table 4.2 displays the mean differences and the corresponding confidence intervals for the mean differences in F1 scores for the classifier trained on all the training data (All) and each of those trained on the smaller datasets (1, 5, 10, 25, 50 utterances per language). On average, using only 1 utterance of English and Muruwari results in a system that is 28 percentage points worse than using all the data (Table 4.2 a). While using 5 or 10 utterances resulted in similar average differences compared to using all the data (10 vs 7 percentage points, respectively), the difference is nearly twice as variable when only 5 utterances per language are used (CI width: 20 percentage points).

Answering our SLI-related questions, then: 1) using 10 utterances per language yields systems whose average performance is within 10 percentage points of using all the data (3892 utterances). 2) a logistic regression classifier suffices for two-way same-speaker SLI using off-the-shelf speech embeddings for SLI (Ravanelli et al., 2021).

4.4 Experiment 2: Automatic Speech Recognition

Recall that for ASR, we seek to answer the following questions: 1) What amount of transcribed speech is sufficient to reliably achieve better than off-the-shelf performance for transcribing Jimmie's Australian English? 2) Using the same amount of transcribed speech, to what extent can ASR system performance be further increased when supplemented with a language model trained on external texts? In this section, we report on experiments conducted in order to answer these questions.

4.4.1 Method

In all our fine-tuning experiments, we fine-tuned the Robust wav2vec 2.0 model over 50 epochs, evaluating every 5 epochs (with an early-stopping patience of 3 evaluations). All training runs started from the same off-the-shelf checkpoint and we kept constant the training hyperparameters, all of which can be inspected in the model training script on GitHub. We varied as the independent variable the amount and samples of data used to fine-tune the model and measured as the dependent variable the word error rate (WER).[7]

In all our experiments, we split the total 81 minutes of transcribed English speech into an 80% training set (65 minutes) and a 20% testing set (16 minutes). The training split of 65 minutes was designated as the 100% training set from which we sampled smaller subsets consisting of 52 minutes (80% of training split), 39 minutes (60% of training split), 26 minutes (40% of training split), 13 minutes (20% of training split), 6.5 minutes (10% of training split), 3.25 minutes (5% of training split), and 0.65 minutes (1% of training split).

We fine-tuned 8 separate models with varying amounts of data and evaluated their performance on the same test set to obtain a first estimate of an amount of data sufficient to achieve better than off-the-shelf performance. We then created 10 new 80/20 training/testing splits for cross-validation in order to establish the variability in WER when only using that minimal amount of data.

We were also interested in whether supplementing the ASR system with a language model further reduced the WER. Our initial labelling work revealed that many errors made by the off-the-shelf system were particularly related to domain- and region-specific English words (e.g. *spear*, *kangaroo*). With permission from the maintainers of the Warlpiri-to-English dictionary, we extracted 8359 English translations from example sentences to obtain in-domain/-region sentences in English, e.g. *The two brothers speared the kangaroo*.

We used this data to train a word-level bigram model using KenLM (Heafield, 2011). While we opted to extract sentences from the Warlpiri-to-English dictionary given it is the largest of its kind for an Australian language, this corpus of sentences still only amounts to 75,425 words (4,377 unique forms), and as such we opted for a bigram model over a more conventional 3- or 4-gram model. With the only change being the inclusion of the language model, we then fine-tuned 10 additional models using the same training and testing splits.

4.4.2 Results

Table 4.3 displays the word error rates (WERs) achieved by a Robust wav2vec 2.0 model fine-tuned with various amounts of transcribed speech. The baseline WER achieved by the off-the-shelf model with no additional fine-tuning is 36.3%. Training with all 65 minutes of data yielded a topline WER of 10.1%. Remarkably, training with less than 1 minute of speech was sufficient to decrease the WER to 19.1%. As a first estimate, the amount of training data that sufficiently improves on the off-the-shelf model appears to be 0.65 minutes of transcribed speech.

[7] Ranging from 0% (best) to 100% (worst), word error rate (WER) is a measure of the accuracy of an ASR system, taking into account substitutions (wrongly predicted words), additions (erroneous extra words) and deletions (missing words).

Training set size	WER	CER
a. 65 minutes (100%)	10.1%	4.2%
b. 52 minutes (80%)	10.1%	4.4%
c. 39 minutes (60%)	11.8%	5.2%
d. 26 minutes (40%)	12.3%	5.5%
e. 13 minutes (20%)	13.2%	6.1%
f. 6.5 minutes (10%)	13.4%	6.1%
g. 3.25 minutes (5%)	15.1%	6.7%
h. 0.65 minutes (1%)	19.1%	8.8%
i. Off-the-shelf (0%)	36.3%	21.5%

Table 4.3: Word and character error rates (WERs, CERs) achieved from fine-tuning the same wav2vec 2.0 model (large-robust-ft-swbd-300h) over 50 epochs using various subsets of data from 65 minutes of Australian English archival audio data.

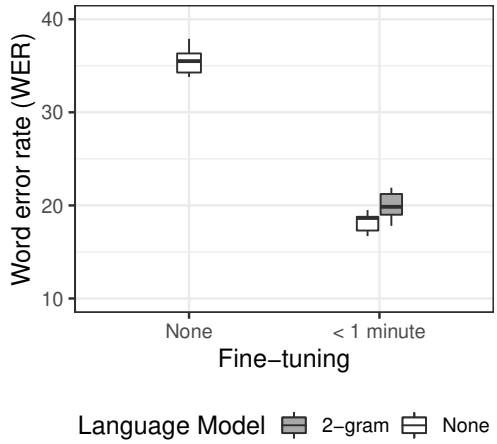


Figure 4.3: Variability in word error rates of training and testing Robust wav2vec 2.0 models over 10 iterations using different samples in the training and testing datasets, holding constant the size of the training set (1% of training set = 0.65 minutes or 39 seconds, on average) and testing set (16 minutes). The off-the-shelf model without fine-tuning was also evaluated on the same 10 testing sets.

To verify that fine-tuning with only 1% of our training data does consistently yield a better than off-the-shelf WER, we conducted cross-validation experiments using 10 additional 80/20 training/testing splits, each time using only 1% of the data from the training split (0.65 minutes or 39 seconds on average). Figure 4.3 displays the results of our cross-validation experiments. First, evaluating the off-the-shelf model on the 10 test sets, we found the baseline mean WER to be 35.6% (standard deviation, SD: 1.48%; range: 33.8–37.9%). The mean WER of the models fine-tuned with only 1% of data and without a language model was found to be 18.2% (SD: 0.99%; range: 16.7–19.5%). These results demonstrate that fine-tuning with less than 1 minute of speech consistently yields better than off-the-shelf performance.

When a bigram language model was used for decoding, we found that the mean WER increased to 20.0% (SD: 1.48%; range: 17.8–21.9%) for the fine-tuned models. These results are inconsistent with our earlier experiments (reported in Appendix A), where we fine-tuned the same off-the-shelf model with 39 minutes of data. In these experiments, decoding with the same bigram model did lead to WER improvements, suggesting that more careful calibration and weighting of the language model may be required in near-zero shot adaptation scenarios.

To answer our ASR-related questions, then: 1) 39 seconds of speech on average is sufficient to achieve a better than off-the-shelf performance for transcribing Jimmie's Australian English speech. 2) the effect on ASR performance of a language model is inconclusive.

4.5 Experiment 3: Timed annotation tasks

In addition to helping provide estimates of the contents of recordings for review by an authorised person, another purpose of this workflow is to help reduce the time required to annotate speech in such a way that excerpts from cleared recordings can be easily extracted for use in relevant initiatives, e.g. creating language learning materials.

The initial process of annotating speech for this purpose involves two tasks: segmentation and transcription, which we illustrate in Figure 4.4 using two clips of Jimmie's speech. In segmentation, the annotator identifies regions of speech and non-speech and also which of the speech regions is English or Muruwari. For a sequence of English sentence fragments such as those in Clip a), the utterances can simply be merged into one. For mixed-language regions such as those in Clip b), separate utterances should be created to allow the Muruwari speech to be easily extracted for use in language learning materials. To create transcriptions for indexing, the annotator transcribes the English segments, given regions segmented and identified as English. We conducted a set of timed annotation tasks to evaluate to what extent the machine-assisted workflow reduces the time taken to perform these two tasks.

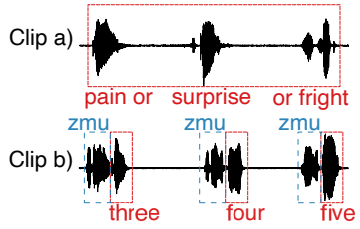


Figure 4.4: Desired annotations for two excerpts of speech from the Jimmie Barker recordings. Clip a) shows a sequence of sentence fragments in English, to be annotated as a single utterance. Clip b) shows alternating Muruwari (zmu) and English speech, to be annotated as 6 utterances.

4.5.1 Method

As detailed in Table 4.4, we gathered for our timed annotation tasks three different recordings approximately 30 minutes in length that were not part of the training and evaluation recordings in the previous experiments. For each timed task, annotators were asked to perform only segmentation or only transcription. For segmentation, they either manually created all time boundaries or corrected machine-derived ones from the VAD and SLI systems. For transcription, they either manually typed in the transcriptions for English speech or corrected machine-derived ones from an ASR system. We tested ASR systems developed earlier in our research (reported in Appendix A), that was fine-tuned on 39 minutes of Jimmy's Australian English speech, and reached a WER/CER of 19/7, as well as a version of the same system augmented with a bigram language model which reached a WER/CER of 14/6. The three recordings and the four annotators and the six annotation tasks were counter-balanced such that each annotator listened to each recording for a given task exactly once.

4.5.2 Results

As detailed in Table 4.4, the segmentation task took 85.5 minutes of work for a 30-minute recording without machine assistance and 82.5 minutes when assisted. That is, correcting time boundaries, inserting missing intervals or removing erroneous ones, and merging/splitting machine-derived segmentations takes nearly the same amount of time as placing these boundaries manually. The waveforms in Figure 4.4 illustrate how the acoustics of alternating Muruwari and English separated by brief pauses look indistinguishable from English sentence fragments separated by similar amounts of pauses — leading to sub-optimal segmentations using a standard, sequential VAD-then-SLI pipeline. The mixed-language nature of this speech may require jointly optimising the VAD and SLI steps.

	Time taken in minutes (Annotator)					
Recording ID (Running time, mins)	**Segmentation only**		**Transcription only**			
	Manual	Assisted VAD+SLI	Manual	Assisted: ASR systems, A–C A	B	C
33-2171A/S1 (31)	88 (A1)	81 (A2)	-	-	54 (A4)	53 (A3)
33-2163A/S1 (33)	83 (A2)	84 (A1)	-	-	57 (A3)	66 (A4)
33-2167B/S2 (32)	-	-	96/87 (A1/A2)	55/71 (A3/A4)	-	-
Mean time taken, in minutes	**85.5**	**82.5**	**91.5**	**63.0**	**55.5**	**59.5**

Table 4.4: Time taken to annotate recordings by four annotators (A1–A4) with and without machine assistance. In the segmentation task, annotators corrected the segmentations by the voice activity detection (VAD) and spoken language identification systems (SLI: trained on 10 utterances per language), or they manually annotated speech regions. In the transcription task, annotators were given intervals of English speech without any accompanying text (manual transcription), or text generated by one of three ASR (A, B C) systems differing in accuracy. System A was an off-the-shelf Robust wav2vec 2.0 model Hsu, Sriram et al., 2021 with no fine-tuning (word error rate/character error rate: 36/22). Systems B (19/7) and C (14/6) were Robust wav2vec 2.0 models fine-tuned on 39 minutes of transcribed English speech, and System C supplemented with a bigram language model trained on external texts.

The transcription task took 91.5 minutes of work for a 30-minute recording without machine assistance and on average 59.3 minutes when assisted (a 35% reduction). We found no meaningful difference between the correction times for transcriptions generated by ASR systems with different levels of accuracy. For transcriptions produced by an off-the-shelf system (WER/CER: 36/22), the correction time was 63 minutes. For systems fine-tuned with 39 minutes of transcribed speech, WER/CER: 19/7 and 14/6, the correction times were 55.5 and 59.5 minutes, respectively.

The closeness in transcription correction times may relate to how an English ASR system whose WER is 30% or less produces good enough transcriptions for editing, according to a crowd-sourced study (Gaur et al., 2016). Here, our transcribers' tolerance for the relatively less accurate off-the-shelf system (WER 36%) may be attributable to their familiarity with the speech domain and speaker (Sperber et al., 2017), having collectively spent nearly 40 hours correcting transcriptions of Jimmie's English by the time we conducted the timed tasks. These results suggest that, where correction is permissible by L1-speaking transcribers of the metalanguage, the time savings over manual transcription could still be gained using an off-the-shelf system that achieves a WER of 30–36% or less for the metalanguage in the recordings.

Nevertheless, we find that the machine-assisted workflow does offer time savings over a fully manual workflow (in line with previous work, e.g.: Sperber et al., 2016, 2017). Specifically, we find that the machine-assisted workflow offers a 20% reduction in overall time to identify regions in the target language and metalanguage and also transcribe the latter, requiring 2.36 hours (82.5 + 59.3 mins) of correction time for a 30-minute recording compared to a fully-manual one which requires 2.95 hours (85.5 + 91.5 mins). Unlike the manual workflow, the fully-automatable workflow can derive first-pass transcriptions to help an authorised person triage recordings.

4.6 Impacts on documentation workflows

4.6.1 Facilitating Muruwari orthography development

A key source of information on Muruwari phonemes and words of interest to the current Muruwari community are two 1969 recordings in which Jimmie Barker discusses an early Muruwari wordlist (Mathews, 1902). This wordlist was created by linguist R.H. Mathews and consists of Muruwari words in his romanisation along with English translations. Using this wordlist, the documentation team is able to shortlist Muruwari words whose romanisation is suggestive of containing sounds of interest (e.g. dental consonants), and then quickly locate in these recordings Jimmie's pronunciation of the words and associated commentary using the time-aligned English transcripts generated for the two recordings.

To give a concrete example, suppose we are interested in finding audio excerpts demonstrating a dental vs. retroflex contrast and based on the wordlist we have shortlisted two candidate words *gadhi* 'uncle' and *gardi* 'bitter' whose romanisations are suggestive of a /t̪/ vs. /ʈ/ contrast (n.b. we use the voiceless symbols here as voicing is not contrastive in Muruwari, as is the case for many Australian languages). Accordingly, for each of these two words, we may search its gloss in the automatically derived transcriptions, arriving at

some results as illustrated in Table 4.5. The searches revealed that Jimmie demonstrates the word for ‘uncle’ at around the 5.10–5.12s region in the recording 04-1313A/S2 and, similarly, demonstrates the word for ‘poison’ at around the 11.42–11.44s region in the recording 10-1706B/S2. As indicated by each region the model has identified as Muruwari (for which I provide IPA transcriptions for illustrative purposes here), we can listen to the neighbouring regions to confirm that the Muruwari forms provided by Jimmie do indeed demonstrate the expected contrast. In this way, the English transcripts provide significantly more streamlined access to untranscribed Muruwari utterances than browsing the recordings in real time.

Recording ID	Timespan	SLI	ASR	IPA
04-1313A/S2	0:05:10.744 – 0:05:11.200	Muruwari		[gaṯi]
	0:05:12.246 – 0:05:12.584	English	uncle	
10-1706B/S2	00:11:42.599 – 00:11:43.594	English	poison	
	00:11:44.708 – 00:11:45.597	Muruwari		[gaṯi]

Table 4.5: An illustration of locating words of interest based on machine-generated transcriptions of a vehicular language, isolated using Spoken Language Identification (SLI) and transcribed using Automatic Speech Recognition (ASR). Note that transcriptions in the International Phonetic Alphabet (IPA) do not exist in the automatic transcriptions and are provided here for illustrative purposes.

Once verified of containing the sounds of interest, the documentation team is able to extract snippets of these words to be included in the community consultation, of which 3 separate workshops have taken place between 2022–2024. In each of these workshops, a subset of sounds are selected for discussion (e.g. dentals and nasals). Based on relevant excerpts found by the documentation team and their suggestions for possible representations, the Muruwari community can collectively discuss and make decisions on questions related to their orthography: do we want both a *t/d* system or write all words with *t* (even when it sounds like a *d*)? how do the words look if we pick one system over the other? Answering such questions move forward the development of the Muruwari orthography and the automated pipeline has helped streamline the process of collating relevant materials for each of these workshops.

4.6.2 Facilitating novel uses requiring information retrieval

While the use of finding words for developing the Muruwari orthography was an anticipated use at the outset of the experiments, we report an unanticipated, novel use that was facilitated by the availability of the English transcriptions. While Muruwari elder Roy Barker was on location at a museum to repatriate an important cultural artefact, he contacted linguists Alison Mount and Jane Simpson to help determine whether Jimmie Barker provided a language word for the artefact and/or discusses its use in the recordings. Due to the availability of the transcriptions, the documentation team was able to provide the requested information with a quick turnaround.

4.7 Conclusion

Many hours of unannotated speech from endangered languages remain in language archives and inaccessible to community members and language learning programs. The time-intensive nature of annotating speech creates one bottleneck, with an additional one occurring for speech in restricted access corpora that authorised community members must vet before annotation can begin. To help alleviate these bottlenecks, I proposed in this chapter an efficient, privacy-preserving workflow for processing recordings where speech in the target language is intermixed with a metalanguage in a more widely-used language such as English. The workflow leverages voice activity detection to identify regions of speech in a recording, and then spoken language identification to isolate speech regions in the metalanguage and transcribes them using automatic speech recognition. The uncorrected transcriptions provide an estimate of the contents of a recording for an authorised person to make initial decisions on whether it can be listened to by those with lower levels of access to correct the transcriptions, which, collectively, help index the corpus.

Chapter 5

Leveraging pre-trained models and supplemental texts to kick-start ASR development

As introduced in the first chapter, the transcription of target languages in language documentation workflows can be incredibly time-consuming, taking 30–50 hours of work to transcribe an hour of recorded speech. Automatic speech recognition (ASR) could potentially speed up this process by helping obtain transcriptions which can then be corrected by humans. However, ASR systems are typically unavailable for many minoritised languages as their development has conventionally required at least tens if not several hundreds of hours of transcribed speech — leaving many minoritised languages in a Catch-22. The recent emergence of large-scale pre-trained models may provide a path forward, given their advertised ability to learn new tasks with little task-specific data. In this chapter and the next, I examine whether we can leverage these models with only limited amounts of transcriptions along with supplemental resources to obtain ASR systems capable of delivering human-correctable first-pass transcriptions. In this chapter, I examine whether 10 minutes of transcriptions may be sufficient to obtain such a system if some text data is also available.

5.1 Background

5.1.1 How does text data help ASR?

As introduced in the second chapter (§2.2.3), text data can be used to derive a language model or lexicon to guide the predictions. A language model helps bias the system to more probable word sequences, e.g. *two dogs* vs. *too dogs*, and a lexicon helps constrain the system output to possible words, e.g. *dogs* vs. **dogz*. Of course, this requires text data in the target language and for some oral languages this may not exist. In the context of language documentation projects, however, there are likely at least various wordlists from which a lexicon can be derived even for such languages. A natural question that arises is whether leveraging text data could help reduce the amount of transcribed speech needed to begin ASR development for documentation workflows and, if so, how much text data is needed?

5.1.2 Related work

An impressive result that has been widely promulgated is that pre-trained wav2vec 2.0 models may make possible ASR system development with only 10 minutes of transcribed speech (Baevski et al., 2020). What is less often reported, however, are the experiment conditions making this result possible: 1) the model was pre-trained on 960 hours of English audiobooks and 2) it leveraged a language model trained on an 803 million word corpus of the very same books (i.e. on an in-domain corpus). A natural question that arises is whether this result is replicable in real-world conditions for languages besides English, where a specialised monolingual model or similarly large amounts of text are not likely to be available.

Compared to previous approaches, it is clear that wav2vec 2.0 models do indeed offer a much reduced upfront cost in terms of transcribed speech. As such, there have been several recent studies appraising the utility of these models for ASR in language documentation settings (e.g. Coto-Solano et al., 2022; Guillaume et al., 2022; Macaire et al., 2022; San et al., 2022; Zhang et al., 2022). Of the studies that did not examine the use of a language model (LM), examining its use is often explicitly mentioned as a clear next step (e.g. Guillaume et al., 2022, p. 176). Of the two studies that have examined the use of a LM with a fine-tuned wav2vec 2.0 model for language documentation projects (Macaire et al., 2022; San et al., 2022), a common theme has been to examine the effect of varying amounts of transcribed speech (e.g. 10–70 minutes) with and without the use of a LM trained on the full text corpus. Similarly, the original wav2vec 2.0 experiments by Baevski et al. (2020) also only examined the use of different amounts of fine-tuning data (10 minutes to 960 hours), with or without the use of LMs trained on the full 803M token corpus.

Given the broader community interest in using fine-tuned wav2vec 2.0 models with LM integration along with the particular appeal that 10 minutes of transcription may be sufficient to kick-start ASR development if some text data can be leveraged, we undertook a series of experiments holding constant the amount of fine-tuning data (i.e. 10 minutes) while varying the amount of LM training text (e.g. 8k–8M tokens) to complement the aforementioned studies.

5.2 Method

5.2.1 Replication and extension on English

We first replicated the original result for English using the wav2vec 2.0 'LS960' model pre-trained on 960 hours of LibriSpeech and the full, in-domain 803M text corpus. For ASR fine-tuning, we used the '`1h/0`' ten minute training set defined in Libri-Light (Kahn, Rivière et al., 2020). Additionally, we extended the experiment by examining the effects of a) reduced amounts of text (8M, 80k, 8k) as well as b) text sourced from an out-of-domain corpus of web-crawled text (Common Crawl-EN: Buck et al., 2014).

5.2.2 Real-world validation on Gronings, Frisian, Besemah, and Nasal

We then conducted validation experiments using data from two pairs of languages differing in the availability of supplemental texts and their representation in the wav2vec 2.0 XLSR-128 model pre-training data (which is 95% European languages). Gronings and Frisian are two West-Germanic languages spoken in the Netherlands and for each language there is an approximately 7.5M corpus of text available (de Vries et al., 2021). We sourced speech data for Gronings from an ongoing documentation project and Frisian data from the FAME! ASR corpus (Yılmaz et al., 2017). Besemah and Nasal are two predominantly oral, Malayo-Polynesian languages spoken in Indonesia and documentation projects' own materials constitute the only available text data (i.e. 9.5k lexica each). For Besemah and Nasal, approximately 45 hours of informal conversations have been collected as part of fieldwork. Part of these collections have been transcribed and are accessible at PARADISEC (McDonnell, 2008, 2019). For all four languages, we constructed comparable 4-hour datasets for our broader set of ASR experiments. Each dataset is composed of an 80/10/10 train/dev/test split, approximately 24 minutes for each of the dev and test sets and 3.2 hours for training set, from which we sampled 10 minutes for the experiments reported here.

5.2.3 Fine-tuning and evaluation procedure

For English, we used the monolingual English wav2vec 2.0 model (Baevski et al., 2020). For the other four languages, we used the multilingual XLSR-128 model (Babu et al., 2022). For fine-tuning these models for the ASR task, we used the HuggingFace Transformers library (Wolf et al., 2020). As introduced previously (§2.2.3), part of the text generation process for ASR involves decoding, which may be assisted using a lexicon or language model (LM) or unassisted, e.g. greedy decoding.[1] We trained each model for 12k steps and then selected for evaluation the best model according to dev set performance.[2] To obtain an estimate of how performance varies with different text data used to derive the lexicon and LM, we conducted the evaluation experiments with 5 different random samples of each size of sub-corpora. All our experiment code is available on GitHub.[3]

5.3 Results and discussion

The results from our experiments are collated in Table 5.1. For each language, E(nglish) – N(asal), the first condition (e.g. E1) reports the baseline ASR performance of a model fine-tuned using 10 minutes of speech using greedy decoding, i.e. without any assistance from a lexicon or language model. The second condition

[1] For decoding with a lexicon and language model, we used beam search decoding via the torchaudio implementation of the Flashlight decoder (formerly wav2letter: Kahn et al., 2022) used in the original paper.

[2] For checkpoint selection, we used greedy decoding (for Besemah and Nasal) or beam search decoding first with fixed parameters (language model weight: 2, word insertion penalty: -1) for checkpoint selection after which we performed a parameter search with the best checkpoint to further optimise the dev set WER.

[3] https://github.com/fauxneticien/w2v2-10min-exps/

wav2vec 2.0 model (Fine-tuning data, 10 mins)	Language model (Text corpus size, no. of tokens)	Mean WER (SD)		Mean CER (SD)	
		dev	test	dev	test
LS-960 (English)	E1. None	40.4	40.2	13.0	12.8
	E2. None, lexicon only (973k)	40.1	40.5	12.3	12.2
	LibriSpeech (in-domain)				
	E3. 4-gram, full corpus (803M)	13.4	14.2	6.51	6.75
	E4. 4-gram, subset (8M)	15.9 (0.09)	16.6 (0.07)	7.41 (0.05)	7.58 (0.07)
	E5. 3-gram, subset (80k)	23.5 (0.13)	24.1 (0.12)	10.5 (0.07)	10.6 (0.10)
	E6. 3-gram, subset (**8k**)	**36.5** (0.20)	**36.8** (0.34)	17.1 (0.09)	17.0 (0.15)
	CommonCrawl-EN (out-of-domain)				
	E8. 3-gram, subset (**80k**)	**27.7** (0.13)	**27.7** (0.10)	12.1 (0.13)	12.0 (0.13)
	E9. 3-gram, subset (8k)	39.8 (0.12)	39.8 (0.24)	18.5 (0.19)	18.3 (0.22)
XLSR-128 (Gronings)	G1. None	43.8	44.0	11.2	11.3
	G2. None, lexicon only (216k)	41.4	40.7	10.5	10.3
	G3. 4-gram, full corpus (7.6M)	23.4 (0.22)	22.3 (0.22)	7.61 (0.62)	7.44 (0.71)
	G4. 3-gram, subset (**80k**)	**35.6** (0.30)	**34.9** (0.51)	11.8 (0.28)	11.4 (0.34)
	G5. 3-gram, subset (8k)	45.6 (0.41)	46.3 (0.24)	16.9 (0.67)	16.7 (0.43)
XLSR-128 (Frisian)	F1. None	55.1	53.1	18.6	18.3
	F2. None, lexicon only (251k)	53.8	51.7	17.9	17.6
	F3. 4-gram, full corpus (7.4M)	36.7	35.2	16.3	15.9
	F4. 3-gram, subset (**80k**)	**44.4** (0.37)	**43.0** (0.22)	20.3 (0.16)	19.9 (0.22)
	F5. 3-gram, subset (8k)	54.2 (0.39)	52.2 (0.29)	26.2 (0.30)	26.0 (0.25)
XLSR-128 (Besemah)	B1. None	62.3	62.1	21.4	21.2
	B2. None, lexicon only (9.5k)	59.9	62.2	20.1	21.2
XLSR-128 (Nasal)	N1. None	67.2	70.7	23.1	25.5
	N2. None, lexicon only (9.5k)	70.1	75.1	24.1	26.1

Table 5.1: Results of fine-tuning wav2vec 2.0 models (LS960: pre-trained on 960 hours of English; XLSR-128 pre-trained on 400k hours of speech from 128 languages) for automatic speech recognition (ASR) using only 10 minutes of transcribed speech from a target language and with varying amounts of reliance on supplementary text data. ASR performance measured by word error rate (WER) and character error rate (CER). For subset experiments, reported means and standard deviations (in parentheses) were derived across 5 runs with different samples of the text corpus. Alphanumeric labels (E1–N1) used to assist with in-text references to results. Highlighted numbers indicate the smallest supplementary text corpus size which yielded an appreciably lower WER than without using a lexicon or language model, i.e. the 'None' condition.

(e.g. E2) reports the ASR performance of the same model when lexicon-assisted decoding is used. The remaining conditions (e.g. E3–E9) report the conditions where decoding was assisted with both a lexicon and language model, derived from either the full text corpus (e.g. E3) or various subsets (e.g. E4–E9). For each language, the result highlighted in bold indicates the text corpus size that yielded an appreciably lower WER than the baseline condition (e.g. E8 when using an out-of-domain text corpus for English).

For English, we found that fine-tuning with 10 minutes of transcribed speech yielded a test set word error rate (WER) of 40.2 (Row E1) without the use of supplementary texts and a WER of 14.2 (Row E3) using a lexicon and language model derived from the full 803M LibriSpeech text corpus. These WERs are consistent with those reported by Baevski et al. (2020, Table 9), respectively: 45.3 and 13.1, allowing for some error likely from differences in the 10 minute training sets.[4]

For all languages, we found that using only a lexicon did not appreciably reduce the WER. For example, for English, the test set WER remained practically the same with or without a lexicon (Row E1 vs. E2: 40.2 vs. 40.5). For Nasal, the test set WER increased from 70.7 without a lexicon (Row N1) to 75.1 when using a small 9.5k token lexicon (Row N2), which led to many erroneous substitutions from the combination of an already high WER and high out-of-vocabulary rate.

For English, Gronings, and Frisian, we found that using lexica and language models derived from 80k of out-of-domain supplementary texts appreciably reduced the mean test set WER. For English, the test set WER was reduced from 40.2 (Row E1) to 27.7 (Row E8). For Gronings, the test set WER was reduced from 44.0 (Row G1) to 34.9 (Row G4). For Frisian, the test set WER was reduced from 53.1 (Row F1) to 43.0 (Row F4).[5] These results suggest that where supplemental texts in the upper tens of thousands or more are available, fine-tuning a wav2vec 2.0 model with just tens of minutes of speech holds promise for deriving first-pass transcriptions near the 20-30% rule-of-thumb (Gaur et al., 2016; Sperber et al., 2016), provided the language is also well-represented in the pre-training data of the model.

For languages under-represented in speech pre-training corpora and for which tens of thousands of words in supplementary texts are not available, we suspect it remains unavoidable for the time being that more than 10 minutes of transcribed speech is required to begin ASR system development. For reducing the out-of-vocabulary rate with lexica and LMs derived from small text corpora, decomposing words into sub-word units (e.g. syllables) may be worth investigating. In trialling an interactive transcription app for Kunwinjku, Le Ferrand et al. (2022b) report that incorporating syllable unigram frequencies derived from a word list improved the word retrieval performance by 4% (F-score). For reducing the performance gap due to under-representation in the pre-training data, continued pre-training may be one viable approach (e.g. Nowakowski et al., 2023).

[4] There was no indication as to which of the six 10-minute Libri-Light training sets were used in the original experiments.

[5] This relatively higher WER for Frisian is likely due to the Frisian corpus consisting of news and radio broadcasts, compared to the read speech in Gronings.

5.4 Conclusion

We investigated the real-world performance obtainable when leveraging various amounts of supplemental text data to help kick-start automatic speech recognition (ASR) system development with only a limited amount of transcribed speech. Our results suggest that fine-tuning a pre-trained transformer model with just 10 minutes of transcribed speech may hold some promise for deriving human-correctable first-pass transcriptions if the language is well-represented in the model pre-training data and if the trained ASR system can incorporate a lexicon and language model derived from a text corpus with at least 80,000 words (i.e. roughly the length of a novel). For under-represented languages with little text data, leveraging self-supervised adaptation methods relying only on target language speech may be more fruitful.

Chapter 6

Leveraging untranscribed speech to adapt pre-trained models for under-represented languages

As in the previous chapter, I continue to examine how to effectively use limited target language transcriptions by leveraging other resources. As introduced previously, the ability to train models using only untranscribed speech through 'pre-training' has enabled speech processing models to leverage hundreds of thousands of hours of data. Yet, as the available large-scale speech datasets are dominated by major European languages, these pre-trained models are not well-adapted for languages under-represented in these datasets, resulting in relatively poorer performance when subsequently trained or 'fine-tuned' for ASR on an under-represented target language. To adapt the pre-trained model for a target language using only untranscribed recordings, I examine the method of 'continued pre-training', which, as its name suggests, involves continuing to train the model in the same manner, but using untranscribed speech from the target language. However, since these models are designed to be trained using considerable quantities of data, continuing to train them using only a comparatively small amount of untranscribed speech may reduce the effectiveness of the process. In this chapter, I examine whether the effectiveness of continued pre-training can be maximised by also drawing upon untranscribed speech from a similar language for which larger quantities are available and also explore how best to select this similar language (henceforth the 'donor' language).

6.1 Background

For developing automatic speech recognition (ASR), 'low resource' languages are typically classified as such based on the availability of transcribed speech. Untranscribed speech or reliable metadata about the language are often assumed to be easily obtainable. This assumption may not hold true for under-described languages with little digital representation. For such languages, I address two questions in this chapter: 1) does leveraging untranscribed speech from a similar, higher-resource 'donor' language for pre-trained model adaptation help improve speech recognition in the target language, and 2) how do we select the best donor?

The first question is of interest as ASR system development with little transcribed speech has become viable with multilingual pre-trained transformer models for speech (e.g. wav2vec 2.0 XLSR-128: Babu et al., 2022). Yet, as most languages are under-represented in the pre-training data, directly fine-tuning these models with transcribed speech for ASR in the target language can yield lower performance compared to their well-represented counterparts (Conneau et al., 2023). While recent studies have shown the effectiveness of continued pre-training to adapt these models using untranscribed speech from the target language (Nowakowski et al., 2023; Paraskevopoulos et al., 2024), they involved using 70–200 hours of target language data. For some languages, it may be quite difficult to source this much speech data — even untranscribed.

The second question is of interest as under-described languages face a similar issue of under- and even *mis*-representation in typological databases such as lang2vec (Littell et al., 2017), which itself draws on other databases (e.g. phonological information from WALS: Haspelmath, 2009). For example, as phonological information for Punjabi is not available in WALS, the feature vector within lang2vec is derived by averaging information from neighbouring languages, and this leads to erroneous similarity scores when such data is used for comparisons. Thus, it is unclear whether existing similarity measures can robustly help select a suitable donor language and, further, to what extent these measures are suitable for predicting ASR performance differences as a result of continued pre-training.

Addressing these two questions, this chapter proceeds as follows. In the remainder of this section (§6.1), I provide an overview of the continued pre-training (CPT) task and highlight certain details about the wav2vec 2.0 architecture that will be relevant to later discussions for developing a suitable, predictive measure. Subsequently (§6.2), I present the results of CPT experiments using Punjabi as a target language and sourcing donor data from 8 other Indic languages. In the following section (§6.3), I motivate, analyse, and validate the Acoustic Token Distribution Similarity (ATDS) — a bottom-up acoustic measure for language similarity between a target and donor. Validation experiments on Punjabi as well as three additional languages (Galician, Iban, Setswana) show that the ATDS precisely predicts target language ASR performance as a result of CPT-based adaption using a mix of target and donor data and outperforms all other measures examined. Finally (§6.4), I summarise the findings of this chapter, making clear their limitations and outlining directions for future research.

6.1.1 Continued pre-training: model adaptation using untranscribed speech

As introduced in the second chapter (§2.1.2), pre-training involves the use of a proxy task which is able to leverage unlabelled data (e.g. untranscribed speech). This allows for models such as wav2vec 2.0 XLSR-128 to be pre-trained at very large scales, e.g. using 436k hours of untranscribed speech as illustrated by Stage 1a in Figure 6.1. However, many languages are under-represented in the standard datasets used for this initial pre-training. As such, directly fine-tuning these models for under-represented target languages (i.e. direct from Stage 1a to 2 for Punjabi in Figure 6.1) results in relatively reduced ASR performance compared to their well-represented, European counterparts.

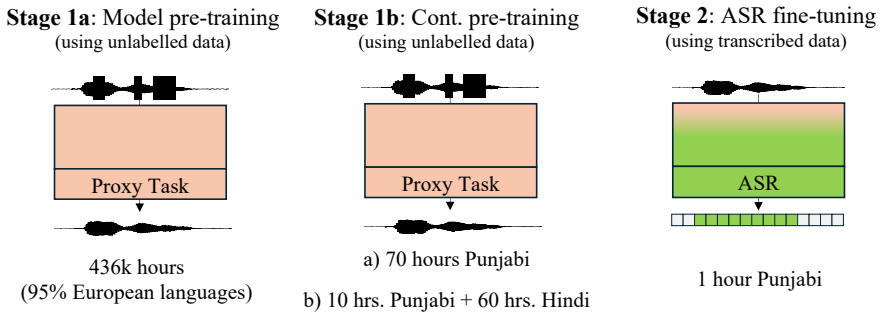


Figure 6.1: Continued pre-training using untranscribed speech from a) large amounts of speech from only the target language (e.g. 70 hours of Punjabi) or b) a mix of the target and a similar, higher-resource 'donor' language (e.g. 10 hours of Punjabi and 60 hours of Hindi).

Given that recording speech is much simpler than transcribing speech, there typically exists far more untranscribed speech than transcribed. Making use of this fact, recent studies (e.g. Nowakowski et al., 2023; Paraskevopoulos et al., 2024) have shown that these models can be adapted for the target language simply by additional or 'continued' pre-training (CPT) using the larger amounts of untranscribed speech, e.g. using 70 hours of Punjabi in Figure 6.1 a). However, such large amounts of speech data may not exist for many digitally under-represented languages. For these languages, we are interested in whether CPT-based adaptation can yield similar levels of improvements when supplementing target language data with data from similar, higher-resource 'donor' language, e.g. using 10 hours of Punjabi with 60 hours of Hindi in Figure 6.1 b).

6.1.2 wav2vec 2.0

As introduced in the second chapter (§2.2), wav2vec 2.0 is a pre-trained model developed for speech processing. We highlight here two important details relevant for our later discussions. The first is that the representations learned by the transformer network are particularly useful for speech applications requiring fine-grained comparisons of acoustic-phonetic content, e.g. speech information retrieval (San et al., 2021), second language pronunciation scoring (Bartelds et al., 2022; Richter & Guðnason, 2023), spoken dialect classification (Bartelds & Wieling, 2022; Guillaume et al., 2023) — particularly those learned by the middle layers of the transformer network (e.g. layers 12–16 of a 24-layer network).

The second detail is that these representations are encoded as vectors in a high-dimensional latent space and emitted at a rate of 49 Hz. Combined with the diversity loss that requires exploration of this space, there is a many-to-many relationship between phonetic categories and these vectors — both in the latent space and in time. For example, as illustrated in Figure 6.2, a single speech sound such as [æ] may last for three time steps (yielding h_2, h_3, h_4). The first two vectors (h_2, h_3) may be very close in the latent space, as they map to the start of [æ] sounds while the third h_4 may map to [æ] sounds preceding [t] and is thus located in a different part of the latent space. Thus to derive phone-like tokens from these representations, we must group them in the latent space (e.g. using k-means clustering) and then again in time according to how the grouped units themselves routinely co-occur (e.g. using subword modelling).

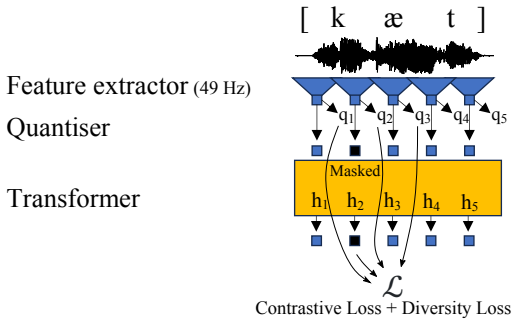


Figure 6.2: Illustration of the wav2vec 2.0 architecture. Adapted from Baevski et al. (2020).

In this way, the goal of tokenisation for both text and speech is driven by the need to derive units of a practical granularity based on the nature of the input: from coarser-grained words to finer-grained sub-words in text and, inversely, finer-grained sub-phones to coarser-grained phones in speech. We return to these two details in our development of the Acoustic Token Distribution Similarity measure for comparing the acoustic-phonetic similarity of two untranscribed speech corpora.

6.2 Experiment 1: A systematic study of pairwise transfer

Given the under-represented nature of most languages in these wav2vec 2.0 models, several studies have investigated continued pre-training (CPT) to adapt them for target languages (e.g. Bartelds et al., 2023; DeHaven & Billa, 2022; Javed et al., 2022; Nowakowski et al., 2023; Paraskevopoulos et al., 2024). Nowakowski et al. (2023) adapted the XLSR-53 model using 200 hours of Ainu. Using the same 40 minutes of transcribed Ainu for ASR fine-tuning, they found that the adapted model resulted in a 8.8% absolute word error rate (WER) decrease over the off-the-shelf XLSR-53 model. For many low resource languages, however, obtaining 200 hours of speech data may not be feasible. In their study of unsupervised domain adaptation for Greek, Paraskevopoulos et al. (2024) found adapting the XLSR-53 model via CPT using a small 12-hour dataset of Greek read speech to be ineffective. However, they found that successful CPT-based adaptation could be achieved with the use of multi-domain data, e.g. 12 hours of read speech mixed with 70 hours of newscasts. Given these findings, we were motivated to investigate whether comparable results could be achieved by supplementing target language data with data from another.

6.2.1 Method

Model training. As our primary interest was in examining how downstream ASR performance is affected by the dataset(s) used in continued pre-training (CPT), we carried out nearly identical experiments as those in Nowakowski et al. (2023) by obtaining their configuration file for wav2vec 2.0 pre-training. Similarly, we follow the official ASR fine-tuning recipe suitable for 1 hour of transcriptions. For both procedures, we made modifications to suit our hardware configuration and compute budget, as detailed in Appendix B.

Data. For a systematic investigation of how downstream ASR performance in a given target language varies with the choice of supplemental language added during CPT, we required a dataset with some specific characteristics: a) contains a variety of both similar and dissimilar languages with, b) at least 60 hours per language, c) and collected in relatively similar acoustic conditions, and d) not have already been used in the original pre-training process. Accordingly, for this set of experiments, we sourced data from IndicSUPERB (Javed et al., 2023), a dataset of 12 Indic languages containing 65–180 hours of read speech per language.

Amongst these 12 IndicSUPERB languages, we selected Punjabi as our target language as it is relatively low-resourced (cf. Hindi, Tamil), its geographical and typological location (yielding a variety of closer and farther geographic/typological distances to the others), and also native speaker-linguist expertise on our research team for data validation and error analysis.

As illustrated in Figure 6.3, we began by selecting for our top-line condition a random 70h subset of Punjabi from the total 136h available in IndicSUPERB, and then from this subset a random 10h selection for our baseline, and again a random 1h subset for fine-tuning. We also selected a random 1h validation and 2h test set both disjoint from each other and any data to be used for pre-training.

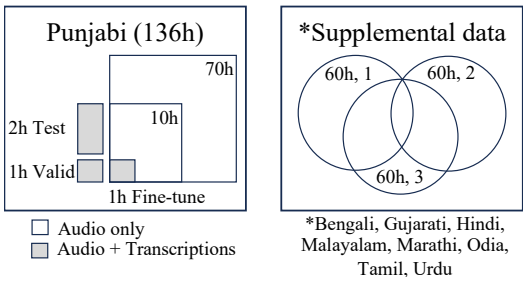


Figure 6.3: Data selection for transfer experiments

Additionally, for each donor language (Bengali, Gujarati, Hindi, Malayalam, Marathi, Odia, Tamil, Urdu), we created three random 60h subsets. Given the total amounts of data available in IndicSUPERB for each language (e.g. 87h for Urdu but 129h for Gujarati), there is however some unavoidable overlap between these subsets for each language (i.e. they are not disjoint 60h splits). Using each of the 60h subsets, we conducted 3 separate CPT runs per language to obtain estimates for both within- and between-donor language differences on downstream ASR performance.

6.2.2 Results and discussion

Compared to directly fine-tuning the XLSR-128 model, adapting the model first via continued pre-training (CPT) with 70 hours of Punjabi yields a large improvement in downstream ASR performance (unadapted 30.8% vs. 22.2% CPT-adapted, 70h). This constitutes an absolute WER difference of 8.6% and is consistent with improvements reported in previous CPT experiments Nowakowski et al., 2023; Paraskevopoulos et al., 2024. We found that model adaptation with only 10 hours of Punjabi still yielded an appreciable improvement over the unadapted model: 5.8% absolute (unadapted 30.8% vs. 25.0% CPT-adapted, 10h).

We now turn to our experiment conditions in which 10h of Punjabi is supplemented with 60h of data from another language. As summarised in Table 6.1, the effects of supplemental language on Punjabi ASR performance can be divided into roughly three strata. In the bottom stratum (E3), we find unrelated Dravidian languages (Tamil, Malayalam) or dissimilar Indo-Aryan languages (Bengali, Odia). When these languages are added as supplemental data during CPT, we find no meaningful difference compared to using only 10 hours of Punjabi (word error rate reduction, WERR: -0.8–0.0%; n.b. higher WERR is better).

In the middle stratum (E2), we find relatively similar Indo-Aryan languages (Marathi, Gujarati, Urdu). When these languages are added as supplemental data, we find modest improvements over using only 10 hours of Punjabi (WERR: 1.6–2.4%). In the top stratum (E1), we find Hindi — the most similar Indo-Aryan language amongst our candidate donors. When Hindi is added as supplemental data, we find a large improvement over using only 10 hours of Punjabi (WERR: 6.0%). In fact, adding the 60h of Hindi results in ASR performance that is in absolute terms close to the 70h Punjabi topline, respectively: 23.5% vs. 22.2% (WER, lower is better).

We have established that there are observable differences in target language ASR performance that vary with the donor language added during continued pre-training and that these differences appear to align with qualitative notions of language similarity. In the next section, we investigate quantitative measures of similarity and evaluate their correlations to these observed differences.

Condition	Test set WER (WERR)		Data for continued pre-training
	Median	Range	
T. In-domain top-line	22.2 (11.2%)	-	70h Punjabi
E1. Most similar	**23.5 (6.0%)**	23.4–23.8	10h Punjabi + 60h Hindi
E2. Similar	24.4 (2.4%)	24.3–24.5	10h Punjabi + 60h Urdu
	24.4 (2.4%)	24.2–24.4	10h Punjabi + 60h Gujarati
	24.6 (1.6%)	24.5–24.7	10h Punjabi + 60h Marathi
B. Only target data baseline	25.0	-	10h Punjabi
E3. Unrelated/dissimilar	25.0 (0.0%)	25.0–25.2	10h Punjabi + 60h Odia
	25.1 (-0.4%)	25.0–25.4	10h Punjabi + 60h Tamil
	25.1 (-0.4%)	25.0–25.3	10h Punjabi + 60h Malayalam
	25.2 (-0.8%)	25.1–25.2	10h Punjabi + 60h Bengali
U. Unadapted XLSR-128	30.8 (-23.2%)	-	-

Table 6.1: Automatic speech recognition (ASR) results from fine-tuning wav2vec 2.0 XLSR-128 Babu et al., 2022 with and without adaptation via continued pre-training (CPT). CPT-adapted models were trained for 10k updates using 70 hours of Punjabi for the topline (T), 10 hours of Punjabi for the baseline (B), and 10 hours of Punjabi combined with 60 hours of data from another language for the experiment conditions (E1, E2, E3). All models were fine-tuned with the same 1 hour of Punjabi data. ASR performance reported in word error rate (WER; lower is better) and word error rate reduction (WERR; higher is better) relative to the 10 hour CPT baseline (B). For each experiment condition, median and range were obtained from 3 CPT runs per language with different donor data in each run.

6.3 Experiment 2: Evaluating predictiveness of similarity measures

A common method for calculating similarities between languages is to use language vectors from the lang2vec database (Littell et al., 2017), which itself draws on other databases (e.g. phonological information from WALS: Haspelmath, 2009). Wu et al. (2021) investigated how well measures based on lang2vec and other data sources correlated with successful transfer learning for ASR. Of the lang2vec similarity metrics, they found that genetic and geographic measures correlated highly with better ASR performance but, surprisingly, inventory and phonological measures did not. Acoustic similarities as derived from embeddings of a pre-trained spoken language identification model were also found to correlate strongly with better ASR performance. In the context of continued pre-training, we questioned whether the to-be-adapted model could be used for this purpose.

Investigating an analogous question in the text domain, Gogoulou et al. (2023) evaluated various measures for predicting transfer characteristics for transformer language models initially pre-trained on one language (e.g. English) and subsequently adapted to another (e.g. Icelandic), and how these characteristics varied according to the distributions of data in the respective language corpora. They propose a novel metric: the Token Distribution Similarity (TDS), which correlated with positive transfer. As illustrated in Figure 6.4 (a), the TDS is derived by 1) using the pre-trained model's tokeniser to process a sample of data from each language, 2) then generating a token frequency vector for each language, and 3) taking the cosine similarity between these two vectors. Given these promising results for predicting positive cross-lingual transfer for continued pre-training on text, an analogous measure for speech could offer the same predictive property.

In order to compute token distribution similarity for two untranscribed speech corpora, we first need to 'tokenise' the corpora. For this purpose, it may be possible to leverage the speech representations extracted using a middle transformer layer (e.g. Layer 12 of 24) of a pre-trained wav2vec 2.0 model such as XLSR-128. As previously highlighted above, these representations are useful for fine-grained comparisons of acoustic-phonetic content and, to make use for these representations for inducing phone-like tokens, they must first be grouped in the high-dimensional latent space and then again in time based on their co-occurrences.

Indeed, wav2seq proposes this very way of inducing pseudo-tokens.[1] As illustrated in Figure 6.4 (b), the first step is using a pre-trained model (e.g. XLSR-128) to extract speech embeddings from a transformer layer. The second step involves training a k-means model to cluster these embeddings (i.e. group similar sounds together). The cluster indices are converted to characters (by a simple Unicode table lookup, e.g.$10 \rightarrow k$, $19 \rightarrow t$, etc.) and then deduplicated. To discover frequently occurring sound sequences, the third step involves using these character strings to train a subword model (e.g. via sentencepiece: Kudo & Richardson, 2018). Once these models are trained on a subset of the corpus, they are used to derive pseudo-tokens for the rest of the corpus and, in our application, also for candidate donor corpora.

Accordingly, we propose the *Acoustic* Token Distribution Similarity (ATDS) measure, which, as illustrated in Figure 6.4 (c), is a straightforward composition of wav2seq (Wu et al., 2023) and Token Distribution

[1] Originally developed for inducing pseudo-tokens for use in a self-supervised pseudo speech recognition task for jointly pre-training a speech encoder and text decoder.

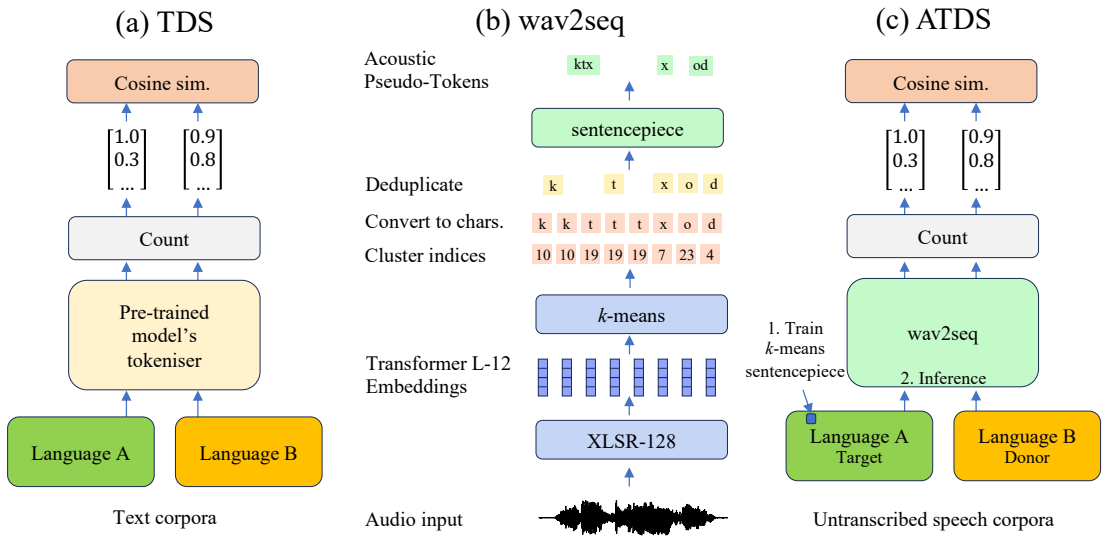


Figure 6.4: Derivation of the Acoustic Token Distribution Similarity (ATDS) measure for predicting positive transfer between two languages resulting from continued pre-training (CPT) of a pre-trained speech model (e.g. XLSR-128). ATDS extends to the speech domain the concept of Token Distribution Similarity (TDS: Gogoulou et al., 2023), shown to predict positive transfer for CPT in the text domain. To account for the token-less nature of untranscribed speech, pseudo-tokens are derived using the wav2seq process proposed by Wu et al. (2023). As is the case for text tokenisation, the goal is to derive units of practical granularity based on the raw input. While typical text tokenisation sub-divides words into sub-words (e.g. *eating* → *eat,ing*), the analogous process for raw speech data involves grouping sub-frames into more phone-like units: first based on featural similarity (e.g. using a clustering method such as k-means) then again based on distributional similarity (e.g. using sub-word modelling via the sentencepiece package).

Similarity (TDS: Gogoulou et al., 2023). In the remainder of this section, we provide 1) a sanity check of the induced pseudo-tokens and 2) two analyses showing that the ATDS between a target language and its candidate donors precisely predicts downstream ASR performance in the target language resulting from continued pre-training of a speech model on target and donor data.

6.3.1 Sanity check of acoustic pseudo-tokens

Method. To discover whether these pseudo-tokens exhibited both within- and cross-language consistency, we conducted an analysis using the Punjabi and Hindi datasets of the CommonVoice corpus (Ardila et al., 2020), for which forced-aligned phoneme labels are available as TextGrids in the VoxCommunis corpus (Ahn & Chodroff, 2022). For this analysis, we trained the *k*-means and subword models on Punjabi (the target language) and then induced pseudo-tokens for both Punjabi and Hindi. We then appended additional tiers containing the time intervals for these pseudo-tokens to TextGrids containing phoneme labels sourced from VoxCommunis and analysed their correspondences. Figure 6.5 illustrates a span of a Punjabi utterance enriched with

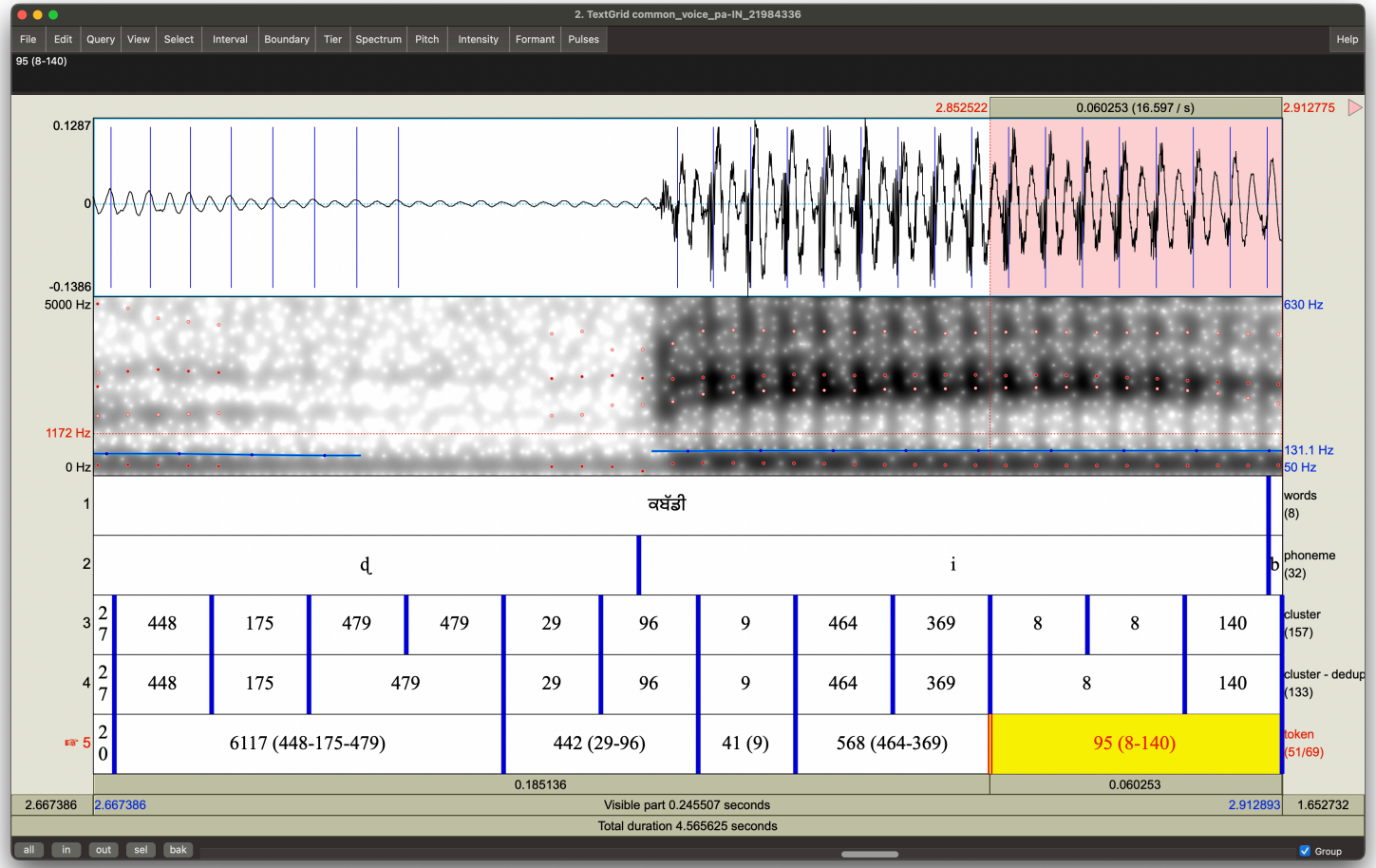


Figure 6.5: Punjabi utterance enriched with pseudo-tokens for analysis. The first two tiers (words, phoneme) are force-aligned annotations sourced from VoxCommunis (Ahn & Chodroff, 2022). The third tier (cluster - 49hz) consists of cluster numbers (0-499) derived by using k-means clustering to group individual time windows based on wav2vec 2.0 embeddings, and the fourth tier (cluster - dedup) retains only uniquely consecutive clusters. The fifth tier (token) further groups unique clusters from the fourth tier based on their frequencies of co-occurrence (e.g. the 95th most frequently occurring span, i.e. token 95, is composed to the speech sounds in cluster 8 and cluster 140).

annotations from the VoxCommunis corpus as well as induced pseudo-tokens. The first tier consists of the Punjabi word and second tier its phonemic representation /ɖi/ derived via an automatic grapheme-to-phoneme conversion process. The third tier (cluster - 49 Hz) comprises of cluster numbers derived by using a k-means model to group together each time window of speech based on their acoustic features, represented using wav2vec 2.0 embeddings extracted from the 12th transformer layer. Given the 49 Hz emission frequency, we can observe that these evenly-spaced units are sub-phonemic and yield repetitions (e.g. '8 8 140' as appears on the right edge of the 'cluster - 49hz' tier).

The task of tokenisation, then, is to derive coarser-grained units, i.e. temporally speaking more phone(me)-like. First, the 'cluster - dedup' tier simply deduplicates repetitions, e.g. '8 8 140' → '8 140'. Then, based on their frequencies of co-occurrence in Punjabi, we further group these clusters into coarser-grained groups, which we term 'tokens'. As indicated on the 'token' tier in Figure 6.5, sequence 8 and 140 was grouped into a single token 95, based on their frequency of co-occurrence in Punjabi. For each token interval, we tallied for analysis the phoneme label present on the phoneme tier at the token midpoint — e.g. the label /i/ for last '95 (8-140)' interval on the token tier in Figure 6.5. Using the models trained on Punjabi, we then repeated the token induction process on the Hindi speech corpus and tallied the correspondences between the tokens and Hindi phoneme labels.

Results. Table 6.2 summarises the top three phoneme labels from Punjabi and Hindi speech corpora associated with induced tokens shown in Figure 6.5. The results show that tokens were broadly associated with a specific speech sound or a set of speech sounds. For example, in Punjabi all occurrences of token 6117 was associated with the retroflex stop /ɖ/ and similarly for token 568 with the high front vowel /i/. These associations were also broadly similar when the tokens were induced on a Hindi corpus and compared to Hindi labels. For example, token 41 is associated with roughly the same set of front vowels in both languages.

Induced Punjabi token ID	Top 3 phoneme labels (% occurrence)	
	Punjabi corpus	Hindi corpus
6117	ɖ (100%)	d (100%)
442	ɖ (78%), i (8%), ʈ (5%)	i (17%), ə (17%), e (13%)
41	i (50%), e (33%), ɪ (7%)	i (54%), e (33%), j (4%)
568	i (100%)	i (72%), j (15%), e (11%)
95	i (90%), ĩ (5%), e (2%)	i (55%), m (7%), n (7%)

Table 6.2: Correspondences of tokens induced on Punjabi and Hindi speech based on frequently occurring sound sequences in Punjabi corpus. Percentages indicate proportion of phoneme label observed at mid-point of induced token interval. Phoneme labels were obtained from grapheme-to-phoneme conversion of language text and time boundaries via forced alignment, available from the VoxCommunis corpus (Ahn & Chodroff, 2022).

There is of course a certain degree of noise in these comparisons, given that all annotations discussed above are machine-derived — from the phoneme labels derived via grapheme-to-phoneme conversion, to time boundaries aligned to speech via forced alignment, to the induced tokens derived based on grouping frequently occurring spans of speech. Thus, there are bound to be certain inconsistencies. For example, while token 442 is commonly associated with /ɖ/, it is almost uniformly split across various vowels in Hindi. Note, however, the occurrence of token 442 Figure 6.5 places it at the boundary between the retroflex and vowel, and such misalignments between automatically-induced tokens and forced-aligned time boundaries may help explain this difference. At the same time, despite such inconsistencies, we are able to observe some broad level of within- and cross-language consistency between tokens and regularly occurring speech sounds in corpora. Given this broad consistency, these tokens may still be useful for measuring how similarly distributed speech sounds are between two corpora.

6.3.2 Comparison of similarity measures

Method. In our first analysis, we examined how well ATDS can account for the results of the Indic language experiments and how ATDS compares to other measures. Accordingly, for ATDS, we trained the relevant wav2seq models on Punjabi (the target language), induced tokens on Punjabi and all donor languages, then calculated the token frequency vectors and computed the pairwise cosine similarities. As in Wu et al. (2021), we also independently computed for the Indic languages similarities using lang2vec and corpus-level means of utterance-level embeddings extracted using a pre-trained spoken language identification model (SpeechBrain: Ravanelli et al., 2021).

Results. This analysis revealed that the two bottom-up acoustic measures provide overall a finer-grained ranking than top-down lang2vec measures. For example, as shown in Table 6.3 Feat. (a combination of all lang2vec data sources), the featural similarity, splits the four languages into two groups: similar (0.6: Hindi–Marathi) and dissimilar (0.5: Bengali–Tamil). Additionally, we also found erroneous, perfect similarity values (1.0), resulting from identical language vectors for the category (e.g. Phon.: Marathi-Punjabi), likely an artefact of data imputation. These results demonstrate that while top-down data may be suitable for more noise-tolerant applications (e.g. large-scale typological comparisons), they may not be well suited for helping select donors for a specific under-described target language if the relevant metadata is unavailable or inaccurate.

While both acoustic measures accurately select Hindi as the most suitable donor, ATDS provides a better ranking of the donor languages. As shown in Table 6.3 (SB), the measure based on SpeechBrain embeddings ranks Gujarati as being as dissimilar to Punjabi as Bengali/Malayalam. We make a similar observation as above that using embeddings from a different pre-trained model than the one to be adapted via CPT risks adding unwanted noise to an inherently hard task. Leveraging the representations of the pre-trained model to be adapted reduces this risk, reflected in ASR improvement being most correlated with ATDS ($r = 0.89$; note that similarity is positively correlated with word error rate reduction, WERR).

	Punjabi (PAN)									
	Donor Lang.	Median WERR (of 3 runs)	*Similarity Measure*							
			ATDS	SB	lang2vec					
					Syn.	Geo.	Feat.	Inv.	Gen.	Phon.
E1.	Hindi	6.0	0.96	0.96	0.67	1.0*	0.6	0.67	0.38	0.41
E2.	Gujarati	2.4	0.93	0.82	0.46	0.9		0.72	0.43	1.0*
	Urdu	2.4	0.93	0.88	0.51			0.67		
	Marathi	1.6	0.92	0.89	0.47			0.65		
E3.	Bengali	-0.8	0.90	0.81			0.5	0.66	0.38	0.38
	Malayalam	-0.4	0.89	0.83	0.32			0.64	0.00	1.0*
	Odia	0.0	0.87	0.71				0.65	0.43	
	Tamil	-0.4	0.86	0.76	0.47			0.59	0.00	
Correlation of measure to WERR:			**0.89**	0.78	0.79	0.77	0.83	0.55	0.48	-0.31

Table 6.3: Acoustic Token Distribution Similarity (ATDS) measure between Punjabi and donor language predicts downstream speech recognition performance as measured by word error rate reduction (WERR; higher is better) when fine-tuning the wav2vec 2.0 XLSR-128 model adapted using continued pre-training (CPT) on 10 hours of target and 60 hours of donor language speech. Other similarity measures for comparison are derived embeddings of the SpeechBrain language identification model and from the lang2vec database (syntactic, geographic, featural, inventory, genetic, and phonological). * indicate erroneous similarity scores resulting from identical, imputed vectors within the database. Correlations (Pearson's *r*) calculated using 24 data points (8 donor languages x 3 CPT runs per language with different donor data in each run).

	Galician (GLG)		Iban (IBA)		Setswana (TSN)	
E1.	SPA (0.96) 10h GLG + 60h SPA	WER (WERR) **13.7 (8.7%)**	ZSM (0.91) 7h IBA + 60h ZSM	WER (WERR) **15.9 (4.2%)**	SOT (0.96) 10h TSN + 56h SOT	WER (WERR) **11.6 (7.9%)**
E2.	POR (0.89) 10h GLG + 60h POR	13.9 (7.3%)	IND (0.88) 7h IBA + 60h IND	16.4 (1.2%)	NSO (0.88) 10h TSN + 56h NSO	12.0 (4.8%)
B.	10h GLG	15.0	7h IBA	16.6	10h TSN	12.6
U.	-	15.4 (-2.7%)	-	21.4 (-28.9%)	-	20.8 (-65.1%)

Table 6.4: Validation of the Acoustic Token Distribution Similarity (ATDS) measure for predicting target language automatic speech recognition (ASR) performance as a result of continued pre-training (CPT) of the wav2vec 2.0 XLSR-128 model using mix target and donor language data. For each target language (Galician, Iban, Setswana), U. indicates ASR performance from using the unadapted XLSR-128 model, B. indicates performance from CPT adaptation with only target language data (7–10 hours), and E1–E2 using target language data supplemented with 56–60 hours of donor language data. Parentheses next to donor language names indicate ATDS to the target language. Percentages within parentheses indicate word error rate reduction (WERR; higher is better), relative the baseline word error rate (B) within the same column. Donor language codes are: Spanish (SPA), Portuguese (POR), Malay (ZSM), Indonesian (IND), Sesotho (SOT), Sepedi (NSO).

6.3.3 Validation of ATDS beyond Indic languages

Method. In our second analysis, we examined whether the ATDS measure generalised beyond the Indic languages through identical CPT experiments on a typologically varied set of target languages. We selected triplets of languages consisting of 1) a target language, 2) a language more similar to the target as measured by ATDS, and 3) another language relatively farther. As summarised in Table 6.4, the target languages were Galician (West-Iberian), Iban (Malayic), and Setswana (Sotho–Tswana). For Galician, ATDS predicted that Spanish (SPA: 0.96) was more similar than Portuguese (POR: 0.89); for Iban, Malay (ZSM: 0.91) more than Indonesian (IND: 0.88); and for Setswana, Sesotho (SOT: 0.96) more than Sepedi (NSO: 0.88).

Results. These CPT experiments are summarised in Table 6.4. We first note the large difference between Galician and the other languages in the improvement yielded by CPT baselines (B) compared to fine-tuning the unadapted XLSR-128 (U). As we sourced Galician data from CommonVoice (on which XLSR-128 was already pre-trained), CPT yields little further gain (U. 15.4% vs. B. 15.0%). By contrast, ASR performance was much improved via CPT adaptation for both Iban (U. 21.4% vs. B. 16.6%) and Setswana (U. 20.8% vs. B. 12.6%). These results constitute further evidence that directly fine-tuning massively multilingual models can yield sub-optimal performance for under-represented languages and that continued pre-training can help close this performance gap.

We found that the ATDS predictions are borne out for all three target languages (even for Galician in spite of relatively reduced benefits). As shown in Table 6.4 for each of the target language columns (Galician, Iban, Setswana), larger improvements in target language ASR performance are observed as a result of continued pre-training on target language data supplemented with data from a more similar language as measured by ATDS than a less similar one (respectively, rows E1. vs. E2). Combined with results for Punjabi above, our findings altogether provide strong evidence for the effectiveness of ATDS for predicting positive transfer between target and donor languages for CPT-based model adaptation.

6.4 Conclusion

For developing automatic speech recognition (ASR) systems for languages with very few resources, I demonstrated that massively multilingual pre-trained models for speech can be successfully adapted via continued pre-training using a mix of data from the target language and supplemental data from a similar, higher-resource 'donor' language. Additionally, I motivated and proposed the Acoustic Token Distribution Similarity (ATDS) — a novel measure of similarity to predict positive transfer between a donor and target language. Across a set of typologically different target languages (Punjabi, Galician, Iban, Setswana), I showed that the ATDS between the target language and its candidate donors precisely predicts target language ASR performance.

This predictive capability of ATDS can be attributed to leveraging the knowledge of the pre-trained model to be adapted, its inductive biases and training objectives, and the distributions within the candidate datasets that will be used to adapt it. It is indeed expected that this measure then is able to predict downstream task improvements better than measures based on other models or external information — the latter of which is

not always available or reliable for under-described languages. Given the high cost associated with continued pre-training, however, I argue that using a well-calibrated, task-specific measure minimises the chance of costly, unexpected outcomes.

These findings, naturally, come with some limitations. To focus on exploring transfer between languages, the standard wav2vec 2.0 pre-training recipe was used. I acknowledge that this requires a large compute budget beyond what is affordable in many low resource scenarios. In future, I hope to investigate whether or to what extent transfer learning can be combined with more compute-efficient adaptation methods. To conduct a systematic study of pairwise transfer, I used domain-matched, high-quality ASR datasets containing mostly read speech and only examined sourcing data from a single donor language. Questions relating to multi-donor and multi-domain transfer, how such interactions affect downstream performance in the target language, as well as experiments with real-world documentation data will need to be addressed in future research.

I make a final observation here that various target-donor language pairs where successful transfer was observed exist in linguistic situations with significant, sustained contact. For example, Galician is a minoritised language in Spain and virtually all Galician speakers are bilingual in Spanish (de la Fuente Iglesias & Pérez Castillejo, 2020). Similarly, Macaire et al. (2022) report that fine-tuning a model pre-trained on French was particularly successful for Gwadloupéyen and Morisien, two French-based creole languages. These findings suggest that to develop truly inclusive speech technologies in a resource efficient manner, what will be required is a nuanced understanding of what factors linguistic and non-linguistic yield sufficiently high levels of cross-lingual similarity which in turn permit positive transfer.

Chapter 7

Conclusion

> Have we addressed the transcription bottleneck? The answer is a qualified yes. [...] We are more selective in where to "transcribe exhaustively." This is how we manage the transcription trap. [...] [This approach] offers diverse workflows and flexibility in the face of diverse language situations. Finally, it opens up new opportunities to engage with speakers. These people offer the deepest knowledge about the language and the best chance of scaling up the work. This is their language.
>
> Bird, S. (2020). Sparse transcription. *Computational Linguistics*, 46(4), 713–744.

The goal of language documentation is to produce a multi-purpose record of a language (Himmelmann, 1998). Given this goal, the recorded materials may encompass a variety of linguistic topics and cultural practices and, consequently, materials appropriate for one purpose (e.g. linguistic analyses) may be entirely inappropriate for others (e.g. creating language learning materials). Thus, the ability to efficiently locate and extract only the relevant information from recorded materials is a need shared among in wide variety of use cases — from linguistic and lexicographical analyses, to development of language materials, to simply wanting to listen to the speech of one's elders.

Plain text searches on transcriptions of the spoken content have been the primary pathway for search and retrieval of material in documentation corpora. Yet as recording speech is much easier than transcribing the recorded speech, many recordings collected for documentation efforts remain untranscribed. This leads to a transcription bottleneck, which in turn limits how easily untranscribed materials can be accessed. There are several contributing factors to the transcription bottleneck — from the need for a writing system and transcribers literate in this system to inefficiencies in the conventional transcription workflow.

Towards alleviating or bypassing this bottleneck altogether, I substantiated the importance of addressing two research questions: first, how to bypass the need for transcription and derive a searchable index, and second, how to make most effective use of available resources to develop automatic speech recognition (ASR) to help accelerate the transcription work. I argued that solutions that address each of these questions can be broadly viewed as diversifying the pathways that can help meet diverse information needs. Leveraging recently developed pre-trained models for speech processing, I examined for each pathway two possible implementations that cater to a common but distinct scenario within language documentation projects.

For scenarios where the speech corpora comprise recordings only in the target language(s), I examined the use of pre-trained models as part of a spoken term detection system. Such systems permit users to perform voice-based searches using spoken queries and in turn permit confirmed hits to act as an index of the corpus (Bird, 2020; Le Ferrand et al., 2020). Evaluated using real world documentation data from 8 languages, results showed that in the worst case scenario using pre-trained models on average can help retrieve 40-50% of all occurrences of a given query. Given that the alternative is scrubbing through untranscribed audio recordings in near real time, this method helps offer *some* form of search and retrieval for uses requiring an exact match. For more discovery-oriented uses (e.g. finding near minimal pairs), however, this method can readily return the most similar sounding near matches (e.g. *halo*, *hollow*, etc. given the query *hello*).

For scenarios where the speech corpora comprise recordings with a more widely spoken vehicular language used for commentary and questions (e.g. *What is the word for 'soft'?*), I examined the use of pre-trained models to automatically isolate and transcribe the vehicular language. The vehicular language transcriptions act as a searchable proxy index, indicating the approximate locations where certain target language words and phrases are being discussed. Evaluated on a 136-hour corpus of Muruwari (an Australian Aboriginal language), results showed that pre-trained models can be readily adapted for this processing pipeline and reduce transcription time by 20%. The pipeline has helped meet information needs not only for anticipated uses such as finding words for developing an orthography but also novel, unanticipated uses such as mentions of a specific cultural artefact for use as evidence in repatriation proceedings.

For scenarios where some external text sources are available, I examined whether leveraging such data can help kick-start the ASR development process when only limited amounts of transcribed speech is available (e.g. 10 minutes). The text data can be used to develop a lexicon which limits ASR system output to possible words (e.g. **dogz* $\rightarrow$ *dogs*) or a language model which helps bias the ASR system to more probable word sequences (e.g. *too dogs* $\rightarrow$ *two dogs*), helping yield better quality transcriptions requiring fewer corrections. This approach was evaluated using two pairs of minoritised languages: Gronings and Frisian (two West-Germanic languages spoken in the Netherlands), and Nasal and Besemah (two Malayo-Polynesian languages spoken in Indonesia). Results showed that while this approached was promising for the two West-Germanic languages, typically over-represented in the data used to develop pre-trained models, ASR performance on the two under-represented Austronesian languages remained poor — suggesting the need for a different approach for such languages.

For scenarios where some untranscribed audio is available, I examined whether this data can be used to

adapt pre-trained models to reduce the ASR performance gap resulting from under-representation. In particular, I examined whether the effectiveness of this adaptation process could be increased through sourcing supplemental data from another similar language, i.e. a donor language. I hypothesised that acoustic-phonetic and phonotactic similarity between the target and donor languages predict the effectiveness (as measured by better ASR performance in the target language) and develop a novel measure to determine this similarity — the Acoustic Token Distribution Similarity (ATDS). Across a set of typologically different target languages (Punjabi, Galician, Iban, Setswana), I show that the ATDS between the target language and its candidate donors precisely predicts target language ASR performance and outperforms all other similarity measures examined. In other words, this approach provides a means of using only untranscribed audio to optimise pre-trained models for developing ASR systems for a given target language, which in turn can help accelerate transcription.

7.1 Limitations and future directions

To ensure findings were as informative as possible for applications in language documentation projects, I sought out data from real-world documentation projects for evaluating all implementations above — except the last study on adapting pre-trained models. In order to systematically study the effects of donor language choice on ASR performance in a target language, I used well-matched ASR datasets with careful read speech recorded in low-noise environments. In practice, there may be many potential mismatches between the target language corpora (e.g. read elicitations from a single speaker) and available donor language corpora (e.g. noisy, multi-speaker conversations). Further, given that the proposed similarity measure is an acoustic measure, its validity in comparing corpora with differing noise levels needs further investigation. A follow-up experiment for example could examine the same target Punjabi data while sourcing acoustically mismatched donor data for Indic languages from the Shrutilipi dataset,[1] containing audio from radio archives.

While the vehicular language isolation and transcription pipeline did examine data sourced from a language documentation project, the single-speaker self-elicitation diverges from the more common multi-speaker settings with typically one person eliciting material from one or more speakers of the target language. Thus, the implementation of an equivalent pipeline will require the use of a speaker diarisation model, which helps differentiate who is speaking when. Additionally, our timed experiments showed that using a cascaded approach where a voice activity detection model (to differentiate speech from silence) feeding into a separate spoken language identification model (to differentiate vehicular from target language speech) yielded poor segmentations that required extensive corrections. Thus, for future implementations for multi-speaker recordings with a vehicular language, a promising approach for jointly performing segmentation and multi-class classification tasks is the recently proposed power-set approach (Plaquet & Bredin, 2023), which for 2 speakers (S1, S2) and 2 languages (L1, L2) could help identify regions of S1L1, S1L2, S2L1, S2L2.

[1] https://ai4bharat.iitm.ac.in/shrutilipi/

Given their recency, the use of large pre-trained models comes with many practical infrastructure challenges which must be overcome before users can leverage them in their existing workflows on their own devices (e.g. a laptop). For example, for use as a feature extractor in a spoken term detection system, the extracted features of excerpts in the corpora would ideally be cached. The efficient storage of these features, however, can require several hundred gigabytes of storage, depending on the size of the speech corpus. Given such costs, all model training and subsequent processing of documentation data using the trained models were conducted by me on a university compute cluster, i.e. not on the machines of the collaborating linguists who collected the data. Fortunately, given global commercial interests in reducing such costs, this current barrier is likely to decrease with time as usage of pre-trained models becomes increasingly ubiquitous. Nevertheless, irrespective of such costs, how best to integrate available technologies into the daily workflow of end users remain a difficult, open question. Some promising avenues towards finding answers are Elpis (Foley et al., 2018), which offers a graphical user interface for ASR training (now including wav2vec 2.0 fine-tuning), and development of plug-ins for using pre-trained models from directly within Elan, e.g. `SileroVAD-Elan` and `xls-r-elan`.[2]

Last, but not least, I acknowledge that all but one of the studies focused on evaluating system performance, which may or may not directly translate to meeting user needs. On one hand, an implemented system may entirely suffice for meeting identified needs and even novel, unanticipated ones — as in the case of finding mentions of a cultural artefact for use in repatriation proceedings. On the other hand, there may be unanticipated barriers that prevent the adoption of a system, requiring design revisions. For example, in piloting the deployment of a spoken term detection or 'word spotting' system for Kunwinjku (an Australian Aboriginal language), Le Ferrand et al. (2022a) found that the very notion of what even constitutes a word placed a barrier to understanding the purpose of the system by speakers of this polysynthetic language. Le Ferrand et al. (2022b) redesigned the system to enable incremental and interactive 'syllable' spotting, which is more naturally in line with how speakers decompose complex Kunwinjku utterances, and which speakers view as a tool for learning and practising literacy via on-screen orthographic representations of the spotted syllables — a use that is in line with the community's priorities for language preservation.

7.2 A final note

Linguists often find themselves explaining to a lay audience the difference between grammatical prescriptivism and descriptivism. The first conjures up under-theorised notions of rigid practices while the second an openness and curiosity about the diverse ways in which language is used, how these ways vary between speakers and languages, and how these ways are constantly changing. Reflecting on the seminal work of Himmelmann (1998), for which we often cite the quintessential purpose of language documentation — to

[2] Respectively: https://github.com/l12maro/SileroVAD-Elan and https://github.com/coxchristopher/xls-r-elan.

create a multi-purpose record of a language —, Himmelmann (2018) points out that the practice of transcription is severely under-theorised despite its central role in language documentation. One effect of this under-theorisation appears to be certain rigid notions of the transcription process, assuming that all purposes require complete and exhaustive transcription. This assumption and notions that underlie it have recently questioned by Bird (2020), who argues that a key part of combating the transcription bottleneck is knowing when transcription is absolutely necessary and, crucially, when it isn't. Of course, it will be no easy feat to establish this collective knowledge. A first step, though, may simply be an openness and curiosity about the diverse ways in which spoken language recordings are used, how these uses vary from one set of users to another as well as language to language, and which uses are best facilitated by which combination of transcription efforts — both human and machine.

Appendix A

Fine-tuning with a re-initialised final layer

In this section, we describe an earlier set of ASR fine-tuning experiments which were analogous to those reported in §4.4, except for the manner in which vocabulary (i.e. character set) was configured. Following recommended fine-tuning practice,[1] we initialised a linear layer whose output size corresponds to set of characters to be predicted (e.g. 'A', 'B', ...) and is derived from the target training dataset. However, this guidance presupposes that the pre-trained model being fine-tuned is one with no prior fine-tuning for ASR on the same language.

Given the size of our available training data (total 65 minutes), we chose to continue to train the Robust wav2vec 2.0 model,[2] already fine-tuned for English ASR on 300 hours of Switchboard (Godfrey et al., 1992). The results of fine-tuning this model using various-sized subsets of our training data is reported below in Table A.1. Notably, fine-tuning with only 13 minutes of data resulted in a significantly worse than off-the-shelf performance (98% vs. 37%, off the shelf). By deriving labels for the linear layer from our training dataset, the label mappings were scrambled (e.g. from Output 4 = 'E' to Output 4 = 'C'), yielding gibberish predictions during initial fine-tuning. Through this fine-tuning process, 39 minutes of training data were required for the model to (re-)learn the appropriate parameters for English ASR.

By contrast, in our experiments reported above in §4.4, we adapted our datasets to match the vocabulary of the tokeniser included with the off-the-shelf model. By doing so, we were able to achieve better than off-the-shelf ASR performance using only 39 seconds of training data.

Yet, unlike those experiments reported above, the addition of a language model to models fine-tuned with a re-initialised vocabulary yielded better performance. As shown in Figure A.1, the mean WER of the models fine-tuned with 39 minutes of data and without a language model was found to be 19.5% (SD: 2.98%; range: 15–23%). When a bigram language model was included, we found that the mean WER decreased to 14% (SD: 2.30%; range: 11–18%). These findings suggest that while the addition of a language model can be beneficial more experimentation is needed to inform best practices for calibrating and/or weighting the language model in near-zero shot learning scenarios.

[1] https://huggingface.co/blog/fine-tune-wav2vec2-english
[2] https://huggingface.co/facebook/wav2vec2-large-robust-ft-swbd-300h

Training set size	WER	CER
a. 65 minutes (100%)	11%	5%
b. 52 minutes (80%)	13%	5%
c. 39 minutes (60%)	16%	6%
d. 26 minutes (40%)	37%	14%
e. 13 minutes (20%)	98%	78%
f. Off-the-shelf (0%)	37%	22%

Table A.1: Word error rates (WERs) achieved from fine-tuning the same wav2vec 2.0 model (large-robust-ft-swbd-300h) over 50 epochs using various subsets of data from 65 minutes of Australian English archival audio data.

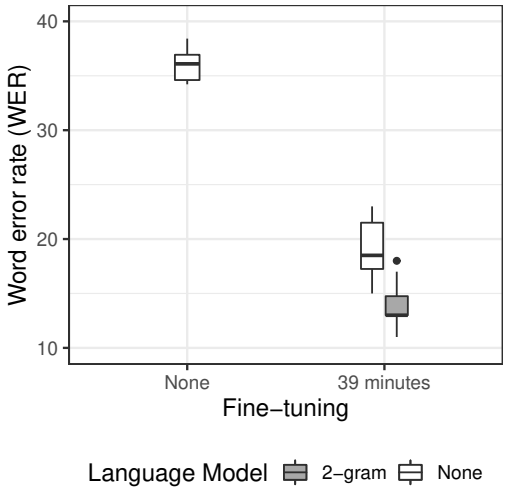


Figure A.1: Variability in word error rates of training and testing Robust wav2vec 2.0 models over 10 iterations using different samples in the training and testing datasets, holding constant the size of the training set (39 minutes) and testing set (16 minutes). The off-the-shelf model without fine-tuning was also evaluated on the same 10 testing sets.

Appendix B

Continued pre-training and evaluation: materials and methods

B.1 Data

Galician, Spanish, Portuguese, and Indonesian data were sourced from CommonVoice (Ardila et al., 2020); Setswana, Sesotho, and Sepedi from NCHLT (Barnard et al., 2014); Malay from MASS (Tan et al., 2009); and Iban from Juan et al. (2015). For Iban only 7 hours of target data was available and for Sesotho/Sepedi only 56 hours per language of donor data was available. Experiments were otherwise identical to the Indic experiments.

B.2 Continued pre-training

We carried out nearly identical experiments as the single-language experiments in Nowakowski et al. (2023) for Ainu, as we obtained their configuration file for wav2vec 2.0 pre-training using the fairseq library.[1] We made appropriate modifications to suit our hardware configuration (4 x A100 40GB GPUs), setting the batch size to 1.5M samples per GPU and gradient accumulation to 16 steps, yielding an effective batch size of 100 minutes.

As in other wav2vec 2.0 multilingual pre-training configurations (Babu et al., 2022; Conneau et al., 2021; Javed et al., 2023), we form multi-lingual batches (specifically, *bi*-lingual in our case).[2] We set our sampling alpha to 0.0, which results in data being drawn uniformly from the two languages (i.e. target is over-sampled). We make this modification based on the CPT method in Paraskevopoulos et al. (2024), where in- and out-of-domain Greek data were evenly sampled in each batch. In this way, we consider this method akin to "similar-language regularisation" (Neubig & Hu, 2018), as we are more concerned with preventing over-fitting rather than learning about the donor data.

[1] https://github.com/facebookresearch/fairseq

[2] Specifically, we use the implementation adapted from https://github.com/AI4Bharat/IndicWav2Vec/

For each CPT run, we start from the official XLSR-128 model checkpoint and update the model for 10k steps. We determined this value based on our pilot runs. We found that 10k updates were sufficient to observe improved downstream ASR performance comparable to previous results e.g. Nowakowski et al., 2023. This choice permitted us to maximise the number of languages compared in this paper, as each run required on average 15 hours (for 10k steps). For select runs, we also verified that no further improvements could be obtained with additional updates (up to 20k).

B.3 ASR fine-tuning and evaluation

For ASR fine-tuning, we follow the official wav2vec 2.0 fine-tuning recipe suitable for 1 hour of transcriptions, modified for our hardware setup. We use a single A6000 48 GB GPU with a batch size of 5.6M samples per GPU and accumulate gradients for 2 steps, yielding an effective batch size of 11.6 minutes. As standard, the feature extractor is kept frozen across all updates and the transformer is frozen for the first 10k of 13k total updates, optimised using a CTC loss. Each fine-tuning run required on average 3.5 hours. For our findings to be applicable for languages with little external text data, we use Viterbi decoding without a language model to obtain transcriptions from the fine-tuned model for evaluation.

B.4 ATDS analyses

For all analyses, we trained the necessary k-means and sentencepiece models on a random 5-hour subset of target language data (except for CommonVoice Punjabi, which had in total 4 hours available in the latest 15.0 release). We adopted hyperparameters based on previous findings: extracting embeddings from the mid-point layer (12 of 24) of the XLSR-128 model e.g. Bartelds et al., 2022; San et al., 2021, and k=500 for k-means and V=10k for the subword model which were reported as optimal values in Wu et al. (2023). Using a 12GB 3060 GPU, embedding extraction required about 10 minutes per data subset. The k-means models trained in about 8 minutes and subword models in less than a minute.

For CommonVoice (CV) Punjabi and Hindi, we conducted similar analyses as those reported by Baevski et al. (2020, Appendix D) for analysing correspondences between the wav2vec 2.0 code vectors and hand-aligned phone labels from TIMIT (Garofolo et al., 1993). In our case we used the labels for CV Punjabi and Hindi via grapheme-to-phoneme conversion, forced-aligned to the audio, and released as Praat TextGrids in the VoxCommunis corpus (Ahn & Chodroff, 2022). We then added tiers containing the wav2seq-induced labels. We hand-inspected several TextGrids for data validation then compiled the correspondences between the wav2seq induced tokens and phoneme labels. We make available all TextGrids as well as the aggregated data.

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